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Simulating Misinformation Propagation in Social Networks: From Linear Chains to Society-Scale Digital Twins

Audited Semantic Degradation in LLM-Agent Networks via
Measurement Sensitivity, Persona Composition, and Network
Conditions

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ABSTRACT

Large language model (LLM) agents can alter factual content when rewriting messages through a network. This thesis transfers a co-authored auditor–node framework to climate and geoengineering texts and audits its uncalibrated claim-recoverability measure, the Misinformation Index (MI).

The campaign combined eight network conditions, 18 persona conditions (12 homogeneous assignments and six heterogeneous mixes), and two auditor regimes. Each configuration propagated six articles: 288 configurations and 1,728 article cells, with one retained run per configuration. Both generator and auditor used `gpt-4o-mini`. Cells with at most one scored event (292) were excluded from propagation means. A stress test used a constructed 63-node hashtag graph.

On 4,237 events identical to their source article, the discrete auditor returned a nonzero score in 38.4 per cent of cases, compared with 1.0 per cent for a continuous assessment of the same texts. This exposes a discrete-scoring defect without validating either regime on altered texts. Across live cells in the campaign that used both auditors, paired assessments of the same generated texts gave mean discrete-minus-continuous gaps of 0.84 in the homogeneous arm and 1.05 in the heterogeneous arm (0–5 scale; equal cell weights). Larger discrete omission penalties contributed 0.32 and 0.61; the residuals combine rubric-dependent and stochastic judge disagreement.

Within each regime, prompts instructing distortion produced higher aggregate scores than prompts instructing preservation. Conspiracy-heavy mixes scored higher than the conspiracy-free mix overall, with article-specific exceptions. Network comparisons combined graph structure, action policy, placement, and scoring coverage; the continuous pooled arm difference changed sign when restricted to runs without logged HTTP errors. On the larger graph, the homogeneous conspiracy assignment scored above the mixed assignment. Its structural check used the input graph itself as a reference and supplied no independent platform validation.

The contribution is a measurement and analysis-population audit of synthetic semantic propagation. Scoring choices changed levels, threshold crossings, and some condition orderings, making instrument checks and explicit scoring populations necessary for interpreting this campaign. The findings describe one model under specified prompts and retained runs; they do not establish human belief change, a causal diversity effect, or online firestorms.

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Abbreviations and Notation

API	Application programming interface.
C1–C5	Descriptive comparison families (Section 4.12).
D-net	Constructed 63-node hashtag graph used for the stress test.
DTFS	Archived composite fidelity diagnostic (Chapter B).
ER	Erdős–Rényi random graph.
H	Homogeneous persona assignment.
He	Heterogeneous persona assignment.
IFD	Information Fidelity and Distortion: the auditor’s item-score record.
LLM	Large language model.
MI	Misinformation Index: uncalibrated claim-recoverability score, 0–5.
MPR	Misinformation Propagation Rate: a mean MI, not a time rate.
RQ	Research question.
SAI	Stratospheric aerosol injection.
SCoPEx	Stratospheric Controlled Perturbation Experiment.
T2c	Continuous-scoring campaign.
T2d	Dual-discrete campaign: discrete headline and continuous sidecar.
k^*	First tick of the terminal scored-tick suffix whose mean MI exceeds 3; a within-window endpoint indicator (Section 4.10).
Live cell	Configuration–article cell with at least two scored events. Cells with at most one are classified as dead.
Cell mean	Event-weighted mean MI within a live cell. Pooled summaries average cells unless stated otherwise.
Node MPR	Mean MI over a node’s scored events; its cell-level companion averages scored-node means.

Chapter 1

Introduction

1.1 Problem and research gap

A scientific proposal can enter public debate as a different message. Harvard’s Stratospheric Controlled Perturbation Experiment (SCoPEX), cancelled in 2024, concerned an aerosol release from a stratospheric balloon (Keutsch Group, 2024). Online discussion nevertheless connected solar geoengineering with chemtrails narratives. Debnath et al. (2023) documented this entanglement across 814,924 English-language geoengineering tweets from 2009 to 2021. Their corpus motivates the substantive case of this thesis, but does not trace a source article through successive paraphrases.

Diffusion research measures how far and how fast content travels. Vosoughi et al. (2018) compared the spread of true and false news, while Goel et al. (2016) distinguished broadcast from viral cascade shapes. A different question concerns whether a reader can still recover the source’s factual claims after several rewrites. This thesis calls the loss of selected source claims *semantic degradation*. It studies that process in synthetic networks of large language model (LLM) agents.

Three strands motivate the design. First, Pfeffer et al. (2014) identify seven factors associated with online firestorms, including network clusters and lack of diversity. Homophily and echo-chamber research also connect composition with exposure (McPherson et al., 2001; Del Vicario et al., 2016; Törnberg, 2018). These accounts do not supply a claim-level measure of text transformation. Second, Debnath et al.’s observed discourse provides substantive themes, but no counterfactual comparison of the same message in differently composed networks. Third, prior co-authored work introduced an auditor–node framework: persona-conditioned agents rewrite messages, and an auditor scores the recoverability of factual items (Maurya et al., 2025). That study used general-news articles, 30-hop linear chains, and a discrete auditor.

The works reviewed in Chapter 2 leave open how auditor design and the population of successfully scored events shape apparent degradation patterns when a fixed-item audit is transferred to climate messages on networks. This bounded gap follows from a targeted review rather than exhaustive coverage. Existing LLM societies and transmission-chain studies establish the feasibility of simulation (Park et al., 2023; Yang et al., 2024; Perez et al., 2025); they do not validate the present measure by transfer.

The measurement problem is central. A reported degradation score depends on the selected questions, the penalty for omission, the judge’s interpretation, and the events included in the average. The same model also writes and judges the messages. Accordingly,

a score difference need not establish a difference in factual content, and the interpretation requires evidence of validity (Cronbach and Meehl, 1955; Messick, 1995; Zheng et al., 2023). This thesis audits the scoring instrument and its analysis population before interpreting composition or network contrasts.

1.2 Aim and research questions

The aim is to extend the inherited framework to climate and geoengineering messages on eight-node networks and establish what its scores support. The main campaign combines eight network conditions, 18 persona conditions (12 homogeneous assignments and six heterogeneous mixes), and two auditor regimes. Its 288 configurations each propagate six curated articles, yielding 1,728 configuration–article cells. Each graph has eight nodes, each article has five audit items, and each run lasts at most eight ticks. One run is retained per configuration. A separate stress test uses a constructed 63-node hashtag graph, D-net. Departures from the proposed empirical corpus and network comparison shifted the executed study towards a descriptive measurement audit (Section C.1).

The analysis uses the Misinformation Index (MI), a 0–5 score in which larger values indicate greater assessed loss or distortion of the selected source claims. H denotes homogeneous persona assignments and He denotes heterogeneous mixes. A live cell has at least two scored events; scorable-cell survival is the share of cells meeting that rule. The within-window threshold indicator, k^* , marks the start of the final sequence of scored ticks with a mean MI above three. Its occurrence therefore describes the observed endpoint, not persistence beyond the run.

Four research questions organise the analysis:

RQ1 (measurement)

How sensitive are claim-recoverability levels, within-window k^* occurrence, and network-condition orderings to the continuous and dual-discrete auditor regimes?

RQ2 (composition)

Within each auditor regime, how do live-cell mean MI and within-window k^* occurrence vary across named persona families and heterogeneous compositions, and how do these patterns compare with the pooled H–He contrast?

RQ3 (network conditions)

How do the eight bundled network conditions differ in same-regime H–He contrasts, live-cell mean MI, and scorable-cell survival within the eight-tick horizon?

RQ4 (stress test)

What can the 63-node constructed hashtag-graph stress test establish about recurring composition patterns, and where does that comparison fail?

RQ1 precedes the other questions because they depend on the scoring instrument. The continuous campaign (T2c) and dual-discrete campaign (T2d) are separate executions; the latter also supplies a continuous sidecar on the same generated events. That sidecar separates same-text disagreement from cross-campaign variation. An unchanged-source control tests whether either instrument assigns degradation to text identical to the source.

The remaining questions describe specified configurations, with scoring coverage and composition reported alongside their means.

1.3 Contributions and scope

The principal contribution is a measurement audit of synthetic semantic propagation. A same-event comparison, an algebraic decomposition, and an unchanged-source control show that assessed claim loss can change without a change in the text being judged. Reconciling summaries with retained histories and examining scoring coverage also show why the analysis population must accompany a headline mean. Together, these analyses establish which interpretations the executed campaign supports and which require independent validation.

Three supporting contributions make that audit possible: a documented climate and geoengineering simulation layer; an analysis by named persona family, mix, and realised exposure; and a mapping of Pfeffer’s factors to the implemented design, accompanied by a structural audit of D-net. The simulation layer includes the six curated stimuli and item sets, persona assignments, network configurations, and run records. The composition analysis distinguishes assigned seats from admitted messages and reports scorable-cell survival alongside conditional severity.

The auditor–node architecture, Misinformation Index (MI), Misinformation Propagation Rate (MPR), and prior general-news chain study are inherited co-authored work (Maurya et al., 2025). They are not claimed as new contributions. The boundary concerns documented research components rather than sole authorship of every current engine routine; Section C.2 records the source provenance.

The thesis studies prompted model behaviour. The personas are author-designed roles, the stimuli are curated summaries, and the graphs are synthetic constructions. Network conditions bundle graph structure with action and assignment rules. One retained run per configuration permits descriptive comparisons, not estimates of repeat-run uncertainty or causal effects of diversity or topology. D-net is not an observed user network and supplies no external validation against Twitter cascades. The simulation recruited no human participants; separate transient processing of public platform records is disclosed in Section 6.7.

1.4 Thesis structure

Chapter 2 reviews the related literature, and Chapter 3 connects it to the measurement and design. Chapter 4 specifies the executed campaign and its departures from the proposal. Chapter 5 answers the four questions, followed by interpretation, validity, ethics, and future work in Chapter 6. Chapter 7 states the contribution within those limits. The appendices retain the audit items, executable design details, structural checks, and robustness results needed to assess the findings; the companion archive preserves the fuller technical record.

Chapter 2

Background and Related Work

The measurement problem in Chapter 1 connects three bodies of work: firestorm and diffusion research, empirical accounts of geoengineering discourse, and LLM simulations of message transformation. This chapter examines what each contributes to the design and how its outcomes were measured or validated. The synthesis in Section 2.9 identifies the bounded gap addressed by the present campaign and motivates the framework in Chapter 3.

2.1 Misinformation as diffusion versus semantic transformation

The misinformation literature distinguishes production, exposure, correction, and belief. Lazer et al. (2018) outlined a research programme spanning these levels, and Vosoughi et al. (2018) treated each of roughly 126,000 fact-checked story cascades on Twitter as a fixed item and measured its diffusion: false stories reached more people, diffused faster, and travelled deeper than true stories. That operationalisation does not quantify source-to-rewrite claim retention, and it concerns diffusion rather than belief, which Lewandowsky et al. (2012) locate in memory (Section 2.7).

Formal diffusion models make the fixed-content assumption explicit. Kermack and McKendrick (1927) moved individuals between susceptible, infected, and recovered compartments at fixed rates; Granovetter (1978) let an individual adopt once the share of adopters around them exceeded a personal threshold; and Kempe et al. (2003) formalised the independent-cascade and linear-threshold models, in which activation passes along edges probabilistically or once the weighted share of a node’s active neighbours exceeds its threshold. These models represent changes in state rather than transformations of text. The rewrite process studied here, in which agents forward, rewrite, or drop a text that an auditor then scores for recoverable source content, cannot be read from activation counts.

Reach is not a single number either: cascade size conceals structure. Goel et al. (2016) examined approximately one billion diffusion events on Twitter, found that events of similar size ranged from single broadcasts to long multigenerational chains, defined structural virality as the mean pairwise distance between nodes in the diffusion tree, and reported that a standard contagion model on scale-free networks did not reproduce the observed variety. Their finding motivates examining realised propagation structure alongside topology labels. In the present campaign, the eight conditions also vary non-topological rules, so their comparisons describe bundled protocols (Section 4.4).

2.2 Online firestorms: theory and operationalisation

Firestorm theory describes the conditions under which negative communication becomes sudden and voluminous, not what the circulating claims say. Pfeffer et al. (2014, pp. 120–124) listed seven interrelated factors: speed and volume of communication, binary choices, network clusters, unrestrained information flow, lack of diversity, cross-media dynamics, and network-triggered decision processes. They offered a conceptual synthesis with three observed cases but neither a generative model nor a scale for the semantic content of firestorm messages. Their claim that diversity among participants would buffer a firestorm is stated by the authors as a hypothesis to be tested in further research, not as a finding.

The following firestorm simulations address conflict, sentiment, and participation rather than recovery of fixed source claims. Hauser et al. (2017) built an agent-based model of conflict diffusion in firm-hosted online communities and compared conflict-management styles; their agents carry opinions and conflict states rather than texts. Lin et al. (2026) introduced CRISP, an LLM multi-agent pipeline evaluated against the sentiment and participation trajectories of four Weibo firestorms; its outcomes are sentiment, attribution, and volume, which this thesis does not measure.

Measurement studies have also developed scales for observed firestorms. Nuortimo et al. (2020) proposed a three-level firestorm scale from media-analytics data, and Nuortimo et al. (2025) proposed and preliminarily evaluated a scale with dimensions of reach, negative-sentiment intensity, and peak duration; neither instrument asks whether the claims within an outrage event remained faithful to their source. Firestorm theory therefore enters this thesis as a design-audit lens rather than as a tested theory, and Section 3.3 maps the seven factors onto what the campaign manipulated, fixed by protocol, measured, or omitted.

2.3 Geoengineering discourse as the substantive case

Solar geoengineering discourse combines scientific uncertainty, governance disputes, and an established conspiracy narrative, which makes it a demanding case for a study of factual drift. Debnath et al. (2023, STAR Methods) analysed 814,924 English-language tweets carrying the geoengineering hashtag between 2009 and 2021 with n-gram frequency analysis, word2vec skip-gram embeddings, lexicon-based emotion scoring, Perspective-API toxicity scoring, and hashtag co-occurrence networks. They found that the chemtrails narrative shaped much of the discussion and that conspiracy themes appeared in political discussions identified through hashtags for the United Kingdom, the United States, India, and Sweden. Those labels identify country-related discourse, not representative national populations. The SCoPEX proposal (Keutsch Group, 2024), discussed in that corpus, supplies the subject of one of the six author-curated articles used here (Section 4.2).

The study is observational and hashtag-based, and its public materials define what this thesis could reuse. The Mendeley deposit (Debnath, 2022a, 2023a) exposes tweet identifiers rather than text, a related OSF deposit (Debnath, 2022b) contains text and

user fields, and the accompanying code (Debnath, 2023b) covers hashtag extraction, embeddings, and emotion and toxicity scoring rather than cascade reconstruction. This thesis attempted no hydration and processed a 5,000-record OSF sample only for hashtag aggregates. D-net is an illustrative synthetic graph inspired by hashtag associations: the sample annotates existing edges but does not generate its topology (Section 4.13). Ruths and Pfeffer (2014) cautioned more generally that platform design, sampling, and automated accounts shape what social media data can show. Without an additional identification strategy, a hashtag corpus does not supply a counterfactual comparison of alternative communication conditions.

The observational account supplies discourse themes for persona design. From the themes reported by Debnath et al., the author constructed profiles for conspiracy, climate action, and environmental concern, then added expert roles. The analysis pools environmental-concern and expert profiles into the science and environment family (Section 4.3). These families are theory-informed prompt groups, not fitted user clusters or estimates of population proportions. The heterogeneous mixes vary their prevalence and placement, so RQ2 compares named compositions alongside the pooled arm labels.

2.4 Topology, homophily, and echo

Network generators define who can expose whom, and three classical generators motivate network conditions used in this thesis. Erdős and Rényi (1960) studied the evolution of random graphs as edges are added uniformly at random; the independent-edge variant used here, $G(n, p)$, connects every pair of nodes with a fixed probability p , so an eight-node realisation requires its own diagnostics rather than a generic low-clustering label. Watts and Strogatz (1998) rewired a ring lattice and showed that sparse random rewiring preserves high clustering while reducing path length, and Barabási and Albert (1999) showed that growth with preferential attachment yields heavy-tailed degree distributions and hubs. Because each configuration here retains one realised eight-node graph bundled with an action policy, the eight network conditions are treated as bundled protocols rather than as contrasts between graph generators (Section 4.4).

Whether a given structure helps or hinders spread depends on the kind of contagion. Centola and Macy (2007) distinguished simple contagion, which a single contact can transmit, from complex contagion, which requires reinforcement from several neighbours. Törnberg (2018) modelled fake news as a complex contagion and found that echo chambers, clustered and homophilous regions of a network, could act as incubators from which a rumour reaches the wider network. These models describe adoption and reinforcement. The present model separately samples forwarding, rewriting, and dropping; using an LLM to rewrite a message establishes neither contagion mechanism. The distinction nevertheless motivates analysing composition jointly with network position.

Homophily links composition to structure. McPherson et al. (2001) reviewed several decades of evidence that similar people connect and concluded that homophily limits the information and attitudes people encounter. Del Vicario et al. (2016) found that Italian Facebook users engaged with either science or conspiracy pages and that the two

communities were polarised, and Newman (2006) formalised modularity, which compares within-community edge density with its expectation under a degree-preserving null model. An echo-chamber label is thus a treatment description rather than a measurement; the D-net stress test therefore reports identity homophily and modularity as observables (Section 4.13), whereas the eight-node echo-chamber, polarised, and hierarchical conditions assign personas to clusters by construction.

2.5 LLM agent societies and persona validity

The question this thesis asks of each generative-agent system is which state it tracks and how it was validated. Park et al. (2022) introduced Social Simulacra, which populate a prototype online community with generated users and posts and were evaluated by designers’ judgements and by participants’ ability to distinguish generated from real subreddit content. Park et al. (2023) then built 25 generative agents with memory, reflection, and planning in a sandbox town and rated believability through interview-style probes. Both systems target believable social behaviour, and neither scores the factual content of the messages that agents exchange.

A second group of systems targets propagation directly. Gao et al. (2023) initialised S^3 agents from real user data and compared simulated information, attitude, and emotion propagation with two observed events. OASIS (Yang et al., 2024) scaled to one million agents and reproduced group polarisation and herd effects reported for human users. AgentSociety (Piao et al., 2025) ran more than 10,000 agents and replicated four social experiments. LAID (Hu et al., 2024) demonstrated LLM decision-making within an agent-based information-propagation simulator. These evaluations concern social behaviour and propagation; they do not report the fixed-source claim-recoverability audit used here.

Two complementary predecessors in Table 2.1 motivate the present combination. Li et al. (2026) drove agents with psychological traits and compared fake-news diffusion across synthetic and empirically derived network structures; their outcome is forwarding, so the study asks who shares an item, not what the item says after sharing. Maurya et al. (2025) built the auditor–node framework this thesis inherits: agents rewrite a general-news article along a linear chain, an LLM auditor answers fixed Boolean questions about each rewrite, and the Misinformation Index counts failed items; they compared persona-homogeneous with persona-heterogeneous chains but did not vary network structure or compare discrete and continuous scoring. The present climate-specific persona prompts are informed by a separate discourse study; their empirical validity remains unestablished.

Persona conditioning raises a validity question that scale does not answer. Argyle et al. (2023) conditioned a language model on demographic backstories and found response distributions close enough to those of the corresponding human subgroups to call the property algorithmic fidelity. Horton et al. (2023) recovered the qualitative results of several classic behavioural experiments from a model treated as a simulated economic agent. Both studies presented these results as evidence for inexpensive probes of theory, not as substitutes for human participants. Larooij and Törnberg (2025) reviewed LLM-based agent modelling, argued that validation is its central unresolved problem, and

listed prompt sensitivity, training-data contamination, stereotype amplification, reduced within-group variance, and compliance with experimenter instructions as threats to any behavioural reading. In this thesis the labels homogeneous and heterogeneous therefore describe prompt conditions, and whether the persona-family gradient reflects simulated disposition or prompt compliance is treated as an alternative explanation in Section 6.6.

2.6 Iterated LLM transmission chains

Serial-reproduction designs follow content through repeated retelling and have recently been run with LLMs. Acerbi and Stubbersfield (2023) replicated five human transmission-chain experiments with ChatGPT, using three summarisation steps per chain and coded retention as the outcome, and observed the same types of content bias as in the human transmission chains: negative, social, threat-related, stereotype-consistent, and counterintuitive material survived preferentially. Perez et al. (2025) ran 50-generation chains across six models, including `gpt-4o-mini`, and found that toxicity, positivity, difficulty, and length converged toward model- and task-specific attractors. Breithaupt et al. (2024) found that ChatGPT summarises once and then changes little, whereas human retellers replace more than half of the words at each step.

These studies distinguish three issues that the present design must keep separate. Retention of selected source items, sentiment change, and false assertion are different outcomes; only the first is scored here. Repeated rewriting can also change a text through attraction under iteration or compression under a length cap, which motivates a neutral-rewrite baseline for a successor design. Finally, correspondence between LLM chains and human transmission requires separate validation. The present campaign examines persona composition and propagation on eight-node networks with a fixed-item auditor, then audits that instrument rather than assuming the earlier evaluations validate it.

2.7 Measurement validity: discrete and continuous instruments

The central dependent variable of this thesis is produced by an instrument, so its meaning has to be argued rather than assumed. Cronbach and Meehl (1955) required that a construct sit in a nomological network of lawful relations to observables before a score can be said to measure it, and Messick (1995) reframed validity as a single evaluative argument about score meaning and the consequences of its use. Neither the prior study nor this thesis has collected human-agreement evidence for MI, so it is treated as an audit score whose interpretation rests on its construction and on the sensitivity analysis that RQ1 reports.

Question-based factuality scoring has a precedent in summarisation evaluation. Fabbri et al. (2022) showed that a question-answering pipeline detects factual inconsistencies in summaries more reliably than lexical-overlap metrics, and Zheng et al. (2023) found that strong LLM judges agreed with human preference judgements at rates comparable to human inter-annotator agreement while documenting position, verbosity, and self-enhancement biases. Neither result validates the auditor used here: it answers fixed

Table 2.1: Generative-agent, firestorm-simulation, and transmission-chain systems reviewed in Sections 2.2 to 2.6. The focal state and the validation evidence are those reported by the cited authors; the last column states how this thesis differs.

System	Focal state or outcome	Scale	Validation evidence reported	Relation to this thesis
Social Simulacra (Park et al., 2022)	Believable community posts and replies	Community prototypes of up to about 1,000 personas	Designer judgements; participants distinguishing generated from real posts	Content believability, not claim recoverability
Generative Agents (Park et al., 2023)	Believable daily behaviour	25 agents	Interview-style believability ratings; ablations	Behavioural believability, no message auditing
S ³ (Gao et al., 2023)	Information, attitude, and emotion propagation	Networks from real user data, two events	Simulated versus observed propagation curves	Aggregate curves; message text not scored
OASIS (Yang et al., 2024)	Platform actions and dynamic following	Up to one million agents	Reproduction of polarisation and herd effects	Scale and platform realism; no claim-level outcome
AgentSociety (Piao et al., 2025)	Urban social behaviour and interaction	More than 10,000 agents	Replication of four social experiments	Population-level effects; no fixed-source claim audit
Li et al. (Li et al., 2026)	Forwarding of fake news under traits	Synthetic and empirically derived networks	Diffusion patterns across structures	Nearest design; outcome is sharing, not content change
LAID (Hu et al., 2024)	Information propagation decisions	Tool demonstration	Demonstration only	Sharing as focal process; no auditor
Hauser et al. (Hauser et al., 2017)	Conflict escalation and opinion under management styles	Agent-based firm-hosted community	Scenario comparison; contingency analysis	Firestorm testbed; agents carry opinions, not texts
Acerbi and Stubbersfield (Acerbi and Stubbersfield, 2023)	Coded content retention along chains	Three-step chains, five experiments, one model	Preregistered replication of human content biases	Nearest process; linear, no personas or graphs, hand-coded retention
Perez et al. (Perez et al., 2025)	Toxicity, positivity, difficulty, and length along chains	50 generations, six models, three tasks	Attractor analysis across models and tasks	Alternative explanation for drift; no factual retention metric
Auditor-node chains (Maurya et al., 2025)	Claim recoverability (MI, MPR)	Linear rewrite chains, general news	Descriptive chain trajectories; homogeneous versus heterogeneous personas	Instrument inherited; this thesis adds climate personas, network conditions, and the regime audit

Boolean items about climate claims, and the same model family generates the rewrites it scores, so the self-enhancement bias becomes a shared generator-and-judge threat. FActScore (Min et al., 2023) decomposes a generation into atomic facts and reports the fraction supported by a reference, and AlignScore (Zha et al., 2023) scores information alignment between any two texts; both target support of the output, whereas MI targets coverage of five selected source claims. MI therefore cannot detect unsupported additions outside the item set and penalises faithful compression that drops an item; its intended construct is recoverability of selected claims, not misinformation in general (Section 3.2).

The choice between discrete and continuous scoring is itself a measurement decision with known costs. MacCallum et al. (2002) showed that cutting a quantitative variable into categories discards information, shrinks effect sizes, and can create spurious relations. Altman and Royston (2006) estimated that a median split of a normally distributed predictor loses efficiency equivalent to discarding about a third of the sample. These warnings motivate reporting sensitivity to the inherited, uncalibrated $MI > 3$ cut that defines within-window k^* .

The two auditor regimes require a further distinction. The discrete prompt does not simply coarsen a continuous judgement: separate calls elicit the two scores and assign different penalties to missing items (Section 4.6). Evidence about dichotomisation therefore does not establish which auditor is more valid. RQ1 treats the regimes as two instruments and checks each against unchanged source text (Section 4.6.1).

A final boundary separates the scored text from the reader. Lewandowsky et al. (2012) reviewed the continued influence effect, in which misinformation shapes reasoning even after people receive and accept a correction, and located that construct in memory and belief. MI records the auditor’s assessment of claim recoverability in a text, not whether any reader accepts the claims. This thesis therefore speaks of claim recoverability, semantic degradation, and factual drift and makes no claim about what readers believe.

2.8 Validation of generative simulation

Simulation methodology distinguishes verification from validation and ties both to the intended use of a model. Sargent (2013) separated data validity, conceptual-model validity, computerised-model verification, and operational validity. Richiardi et al. (2006) proposed a common protocol for agent-based social simulation that requires a transparent model description, sensitivity analysis, replication across random seeds, and a validation test matched to the claim. Windrum et al. (2007) compared three routes to empirical validation and concluded that none removes the identification problems of models with numerous free parameters and few observed series. Larooij and Törnberg (2025) found that the LLM-driven studies they reviewed mostly validated on the face plausibility of agent output rather than on out-of-sample regularities.

The present thesis meets the protocol only in part. The campaign retains one run per configuration and uses unseeded randomness for several generators, which forgoes the replication that Richiardi et al. required (Section 4.7). The D-net stress test checks implementation consistency against its own constructed input graph; it supplies no

independent evidence of operational validity (Section 4.13). Fluent prose alone establishes neither the mechanism nor its correspondence with human behaviour. The design supports an audit of prompt-conditioned synthetic outcomes, not an estimate of prevalence or effects among human users.

2.9 Synthesis

The reviewed studies distinguish diffusion reach, changes in transmitted content, and the validity of the scores used to assess that content. They leave unresolved how scoring rules and selective audit coverage affect apparent degradation when fixed-item audits are transferred to climate messages on networks. This problem motivates three requirements: separate reach from claim transformation, compare auditor judgements on identical texts, and report which events enter each summary. The campaign addresses these requirements through paired scoring, an unchanged-source control, and population restrictions. Chapter 3 defines the quantities and comparisons that this descriptive audit can support.

Chapter 3

Theoretical Framework

3.1 Composition, exposure, and transformation

The framework separates three components: what text an agent receives, how its persona transforms that text, and how an auditor evaluates the output. Composition and network structure jointly shape the first two components. The auditor observes their outputs through a selected item set; its score is an additional model judgement.

Let X_e^{in} and X_e^{out} denote the incoming and outgoing texts of event e at node i , and let P_i be that node’s persona prompt. A successful model rewrite produces $X_e^{\text{out}} = G(X_e^{\text{in}}, P_i; \theta, \epsilon_e)$, where θ denotes the language model and ϵ_e its generation randomness. A forward preserves the incoming text, while a drop produces no outgoing text for that event. Rewrite failures can also preserve the input (Section 4.11). Adjacency, trust, inbox admission, and the stopping rule determine which possible events occur. The outcome therefore depends jointly on persona placement and exposure.

Persona prompts operationalise interpretive tendencies rather than observed beliefs. The present conspiracy prompts explicitly instruct omission or inversion of some audited claims; other prompts instruct preservation. A family contrast can consequently establish prompt responsiveness, but it cannot independently establish that real communities reason similarly. Diversity is likewise multidimensional: changing a heterogeneous mix changes identities, family prevalence, and placement together.

Network structure supplies opportunities for transmission. Hubs, bridges, and clusters may change reach or repeated exposure, as diffusion and complex-contagion research suggest (Goel et al., 2016; Centola and Macy, 2007). Repeated exposure is not itself persuasion in this simulation: agents do not update beliefs. Identity homophily (McPherson et al., 2001) and repeated content are also distinct. A homogeneous assignment has identity homophily one by construction, without demonstrating an emergent echo chamber.

Two outcome dimensions follow. Communication escalation concerns reach, breadth, depth, and timing; content degradation concerns the recoverability of selected claims. Wide diffusion can preserve those claims, while severe local distortion may spread little. Scorable-cell survival is reported separately because it depends on both propagation and successful auditing. No one of these quantities identifies an online firestorm, which also requires sudden communication volume and negative affect (Pfeffer et al., 2014).

3.2 Measurement model and its limits

The inherited framework scores departures from a fixed set of source claims (Maurya et al., 2025). This thesis uses five Boolean audit items per article. For discrete item loss $z_{ej} \in \{0, 1\}$ and continuous item loss $\ell_{ej} \in [0, 1]$, the event scores are

$$\text{MI}_e^{(d)} = \sum_{j=1}^5 z_{ej}, \quad \text{MI}_e^{(c)} = \sum_{j=1}^5 \ell_{ej}. \quad (3.1)$$

Both range from zero to five, but the penalty for omission differs. Equal nominal ranges do not make the scales commensurate. Section 4.6 specifies their rubrics and decomposes their disagreement.

The construct is narrower than misinformation. The items test claim recoverability from a particular source, rather than intent, belief, toxicity, or all possible falsehoods. A hypothetical rewrite could preserve all five keyed answers and append an unrelated false claim; the fixed-item instrument could still assign $\text{MI} = 0$. This is a coverage limit of the measure, not an observed finding about the campaign. Independent assessment would be needed to detect such additions systematically.

Aggregation also changes the question. A node’s MPR averages its scored events; a cell’s mean MI averages all scored events for one configuration and article. Higher-level means then average cells. Despite the inherited word *rate*, MPR is a mean severity score, not change per unit time. No unique branch denominator exists on a graph, so every summary must identify its unit and weighting rule.

For each scored event, the auditor can be represented as $\widehat{M}_{e,r} = J_r(X_e^{\text{out}}, Q, G_Q)$, where J_r denotes the stochastic scoring procedure under regime r , Q the questions, and G_Q their expected answers. Different regimes can disagree on identical text. The disagreement combines coding differences, rubric interpretation, and stochastic judge variation. It reveals measurement sensitivity without identifying the correct judge. LLM-as-judge research and question-based factuality metrics motivate this audit (Zheng et al., 2023; Fabbri et al., 2022); they do not establish criterion validity for these five-item instruments.

An unchanged-source control supplies one necessary check: a text identical to the source should receive zero claim-loss score. Passing that check would still not establish sensitivity to altered text. Measurement validity requires evidence for the intended inference, rather than a plausible name for the index (Cronbach and Meehl, 1955; Messick, 1995).

The threshold $\text{MI} > 3$ is an inherited operational choice. The prior ten-item study used this cut, and the present five-item design did not recalibrate it. The statistic k^* marks the first tick of a terminal sequence of scored ticks whose mean scores exceed that cut. It exists exactly when the mean score at the final scored tick exceeds three. It is therefore a window-specific endpoint indicator, not evidence of enduring degradation or firestorm onset. Dichotomisation also discards score information (MacCallum et al., 2002).

3.3 Pfeffer’s factors in the executed design

Pfeffer et al. (2014) describe seven interrelated factors associated with online firestorms. Their framework supplies a lens for auditing the design, rather than a claim-loss algorithm or a hypothesis tested by this campaign. Table 3.1 identifies what each factor corresponds to here. The design varies bundled proxies for network clusters and lack of diversity, represents aspects of information flow, and fixes the seeding schedule and horizon. It neither measures negative affect nor analyses a communication-volume outcome.

Binary choice in Pfeffer et al.’s account concerns the either–or decision to transmit without discursive interaction. The simulator’s forward, rewrite, and drop actions contain a transmit decision, but the experiment does not vary this factor independently. Similarly, repeated neighbour exposure is not a measured sequence of knowledge, persuasion, propagation, and affirmation. These distinctions prevent a network simulation from being presented as a test of all seven factors merely because it contains connected agents.

3.4 Expectations and identification requirements

Two expectations can be examined descriptively. First, prompts that instruct distortion should yield higher scores than prompts that instruct preservation; named families and mixes should expose variation concealed by pooled arm means (RQ2). Second, changing the auditor should change levels, threshold status, or condition orderings, including on identical texts (RQ1). Neither expectation requires interpreting the scores as human beliefs.

Three further mechanisms require a stronger design. A position effect needs seat-matched assignments with family shares held constant. A cluster–echo effect needs graph structure and persona placement to vary independently. A reach–severity comparison needs common action policies and replicated graphs, with propagation and scoring coverage measured separately. The present bundled conditions cannot identify these effects. RQ3 instead compares their realised outcomes, while RQ4 asks whether a directional composition pattern recurs on one additional constructed graph.

The resulting scope is explicit: the campaign can document model behaviour under specified prompts and instruments. It cannot estimate population effects, establish a general topology ordering, or validate a model of real firestorms. Chapter 4 specifies the operational choices behind that boundary.

Table 3.1: Seven firestorm factors of Pfeffer et al. (2014) and their treatment in this thesis. Proxy varied denotes bundled operational variation; fixed refers to input settings, not realised outcomes; measured denotes descriptive diagnostics; partly represented marks a limited operational analogue. Not separately manipulated distinguishes a represented decision from an independent factor test; not represented marks a missing mechanism.

Factor	Treatment in this thesis	Evidence the design can supply
Speed and volume of communication	Fixed by protocol: single-node seed at tick 0 (drip), eight-tick horizon, no shock condition; realised message volume and timing vary across runs and are not analysed as outcomes.	Reach within eight ticks only; no volume series, half-life, or conversion of ticks to clock time.
Binary choices	Not separately manipulated: all conditions allow forwarding, reinterpretation, and dropping, while their action probabilities vary with the bundled network policy; no binary-action ablation.	No isolated binary-choice comparison; changes in transmission probability accompany other protocol changes.
Network clusters	Proxy varied through bundled network conditions (small world, echo chamber, polarised); measured on D-net as transitivity, local clustering, and modularity.	Condition contrasts under a bundled protocol; structural diagnostics of a constructed graph; no evidence on emergent clustering.
Unrestrained information flow	Partly represented: scale-free hubs and dense echo-chamber blocks on eight nodes, not hundreds of weak ties.	Bundled-condition score and scorable-survival contrasts; no isolated hub-reach or weak-tie test.
Lack of diversity	Proxy varied: homogeneous versus heterogeneous composition, named families and mixes; measured on D-net as identity homophily (0.84 mixed, 1.00 homogeneous by construction).	Composition-conditioned differences in live-cell mean MI and k^* occurrence; a homophily of 1.00 is tautological.
Cross-media dynamics	Not represented as a factor: no legacy-media or broadcast layer; the hierarchical generator is not a substitute.	None.
Network-triggered decision processes	Not represented as a factor: repeated neighbour exposure occurs, but the knowledge, persuasion, propagation, and affirmation stages are not logged.	Exposure counts only; no persuasion or adoption outcome.

Chapter 4

Data and Methods

The study comprises a 1,728-cell campaign and a separate 63-node stress test. This chapter defines the inputs, propagation and scoring rules, retained-run population, and descriptive comparisons. The companion archive supplies full configurations and event histories; the appendices retain the material needed to audit the central claims.

4.1 Prior-work boundary and system overview

The simulation engine, the Misinformation Index, the Misinformation Propagation Rate, and the general-news chain study are prior co-authored work and are not claimed by this thesis. Maurya et al. (2025) introduced the auditor–node architecture, in which persona-conditioned language-model agents rewrite a message and a separate auditor scores each rewrite against ground-truth items, and defined MI as the event score and MPR as its aggregate. The thesis layer, which is the contribution claimed here (Section 1.3), consists of the climate and geoengineering corpus, the persona library and its mixes, the 288 machine-readable configurations, the executed campaign, the parser that turns event records into cells, the analysis scripts and figures, and the text; the two background colours in Figure 4.1 carry this boundary through the chapter.

The executed pipeline moves one article through a network of eight persona-conditioned agents and then through an auditor, and the parser turns the audited events into cells. In Figure 4.1 an article enters the seed node at tick zero; over eight synchronous ticks every receiving node forwards, rewrites, or drops it; the auditor attempts to score each forwarded or rewritten text against the article’s five Boolean items under one of two regimes; and the parser aggregates the event scores into a live-cell mean MI, a within-window k^* , and a live or dead status per configuration–article cell. The following sections follow that order.

4.2 Articles and ground truth

Every cell in the main grid seeds one of six author-curated, source-informed articles, each of which carries five Boolean audit items. Each record holds a title, a source text of roughly 170 to 230 words written for the campaign, five yes-or-no questions, the expected Boolean answer to each, and source notes. The expected answers mix affirmations and denials, so a rewrite cannot score well merely by agreeing with every statement it

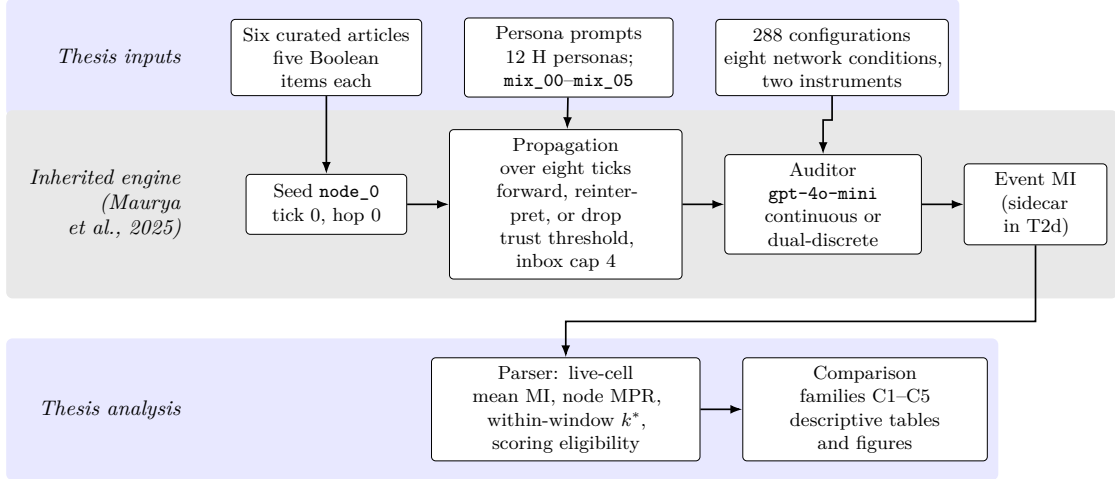


Figure 4.1: Executed pipeline from article to comparison family. The grey band marks the inherited auditor–node engine; the blue bands mark the layer produced for this thesis, which supplies the inputs above the engine and reads the event records below it.

meets. The archive contains no record of independent subject-specialist validation. The supporting spans in Table A.1 make the intended answers inspectable against the curated stimulus text; they do not establish external factual validity. Table 4.1 lists each article’s topic and sources; the checked-in JSON text, not the sources, is the executable stimulus. The merged article library holds 24 records, but the other 18 do not enter the 1,728-cell grid.

The corpus was chosen so that the personas and the articles address the same discourse. Three articles concern the geoengineering debate that Debnath et al. (2023) analysed on Twitter, and three concern established climate-science claims outside it, so a persona’s treatment of geoengineering can be compared with its treatment of the Paris Agreement, the anthropogenic consensus, or polar-bear habitat. Because each article bundles a topic, a wording, and one selection of five items, article identity is a bundled factor, and Chapter 5 reads article contrasts accordingly.

Each configuration processes its six articles one after another from a clean state. Before an article is seeded, the engine clears that article’s history and every inbox, then places the full source text in the seed node at tick zero with hop count zero. No competing article, intervention, or bot module was enabled in any Phase 2 configuration.

4.3 Persona conditions

The profiles are author-designed prompts informed by discourse themes in Debnath et al. (2023), with additional expert roles. They are not fitted user clusters or replicas of observed accounts. The homogeneous arm (H) places one persona at all eight seats and covers 12 profiles: four conspiracy, three climate-action, and five science and environment profiles. The last family combines three environmental-concern profiles with two experts.

Table 4.1: The six campaign articles, their topics, and the principal sources used to author each summary and its five-item Boolean answer key. Three articles address the geoengineering discourse studied by Debnath et al. and three address established climate-science claims.

Identifier	Topic	Principal sources
scopex_2017	Harvard SCoPEX as a kilogram-scale stratospheric measurement experiment, not a deployment	(Keutsch Group, 2024); 2017 volume peak in (Debnath et al., 2023)
chemtrails_gates_2018_2021	Chemtrails, SCoPEX, and Bill Gates in the public scientific record	Discourse analysis (Debnath et al., 2023); contrail physics (Federal Aviation Administration, n.d.)
sai_geoengineering	Stratospheric aerosol injection as a proposed, not deployed, method	IPCC WG I, Section 4.6.3.3 and Box TS.8 (Intergovernmental Panel on Climate Change, 2021)
paris_agreement	The 2015 Paris Agreement and nationally determined contributions	Treaty Articles 2 and 4 (United Nations Framework Convention on Climate Change, 2015)
climate_consensus	Multiple lines of evidence for human-caused warming	(Intergovernmental Panel on Climate Change, 2021); consensus survey (Cook et al., 2013)
polar_bears	Sea-ice loss as a habitat threat despite stable subpopulations	IPCC WG I Arctic assessment (Intergovernmental Panel on Climate Change, 2021); IUCN specialist assessment (IUCN Polar Bear Specialist Group, n.d.)

Complete prompts and the broader library remain in the companion archive.

The prompts partly prescribe the outcome. For example, `conspiracy_believer` instructs the model to invert or omit claims about ordinary contrails, the SCoPEX material, and experiment cancellation. Each conspiracy prompt contains a comparable instruction. The other homogeneous prompts instruct preservation; `climate_scientist` explicitly asks the model to retain facts and quantitative caveats. Family differences therefore function as a prompt-responsiveness check, not evidence of naturally emerging differences between human groups.

The heterogeneous arm (He) declares six ordered mixes of eight identifiers. Their declared conspiracy-seat counts are 4, 2, 0, 1, 2, and 2 for `mix_00` through `mix_05`. Only five generators seat these lists sequentially. The echo-chamber, polarised, and hierarchical conditions instead use two assignment pools, conspiracy and other, alternating by node parity. With a nonempty conspiracy pool, the realised mix places conspiracy profiles at all four even seats. This increases the declared conspiracy-seat counts in `mix_01` and `mix_03–mix_05`. The count remains four in `mix_00` and zero in `mix_02`, whose conspiracy pool is empty. Table A.5 records the declared and realised orders.

The polarised graph’s blocks are contiguous halves, while assignment alternates by parity, so each block contains both pool types. Across the full He grid, conspiracy seats account for 115/384 (29.9 per cent), compared with 11/48 (22.9 per cent) in the declared mixes and 33.3 per cent in H. These are seat shares, not shares of admitted or scored events. The H–He contrast therefore changes identity, prevalence, diversity, placement, and sometimes assignment procedure together.

4.4 Network conditions

Every main-grid graph has a directed graph with eight nodes produced by one of the eight generators in Table 4.2. The linear chain retains the structure of the prior chain study; the ring closes it into a cycle; the Erdős–Rényi (ER) generator samples each ordered pair independently (Erdős and Rényi, 1960); the small-world generator rewires a directed ring lattice (Watts and Strogatz, 1998); the scale-free generator grows a graph by preferential attachment (Barabási and Albert, 1999); and the echo-chamber, polarised, and hierarchical generators are bespoke constructions whose names describe a wiring rule, not a validated platform model. Figure 4.2 shows retained realisations; Table A.4 specifies the generator parameters.

Initial edge trust depends on the generator and, for five generators, on persona similarity. The chain and ring set every edge to 0.70, and the Erdős–Rényi generator draws each trust uniformly between 0.30 and 0.80. The other five generators apply a homophily adjustment: an edge keeps base trust b when the persona map is absent, the persona identifiers agree, or both tag sets are empty. Otherwise it takes $\text{clip}(b - 0.20 + 0.50 J, 0.05, 0.95)$, where J is the Jaccard overlap of the two personas’ tag sets (cf. McPherson et al., 2001). Homogeneous configurations therefore keep base trust everywhere, whereas heterogeneous configurations vary trust with the seated pair, a further channel through which composition and network condition interact.

Table 4.2: The eight network conditions: generator, construction, configured parameters, and the node action policy bundled with each. The chain and ring conditions share a rewrite-heavy policy and a lower trust threshold τ , so contrasts between them and the other six conditions are protocol contrasts, not differences attributable to the generator alone. The seed does not process the ring’s returning message within eight ticks.

Condition	Construction	Parameters	Action policy (forward, reinterpret, drop)	τ
Linear chain	Directed path $v_i \rightarrow v_{i+1}$	edge trust 0.70	0.10, 0.85, 0.05	0.08
Ring	Directed cycle $v_i \rightarrow v_{(i+1) \bmod 8}$	edge trust 0.70	0.10, 0.85, 0.05	0.08
Random ER	Each ordered pair sampled independently; unseeded	$p = 0.42$; trust $0.30 + U(0, 0.50)$	0.35, 0.50, 0.15	0.15
Small world	Directed ring lattice with random rewiring; unseeded	$k = 4, \beta = 0.10$	0.35, 0.50, 0.15	0.15
Scale-free	Preferential attachment from a complete seed of $m + 1$ nodes; unseeded	$m = 2$	0.35, 0.50, 0.15	0.15
Echo chamber	Two chambers by seat parity; within- and between-chamber edge probabilities; seed out-degree at least 2; seeded RNG	probabilities 0.75, 0.08; base trust 0.85, 0.15	0.35, 0.50, 0.15	0.15
Polarised	Two contiguous halves; within- and between-half edge probabilities; seed out-degree at least 2; seeded RNG	probabilities 0.75, 0.05; base trust 0.88, 0.10	0.35, 0.50, 0.15	0.15
Hierarchical	Rooted 3-ary tree with downward and upward edges	down trust 0.85; up trust 0.30	0.35, 0.50, 0.15	0.15

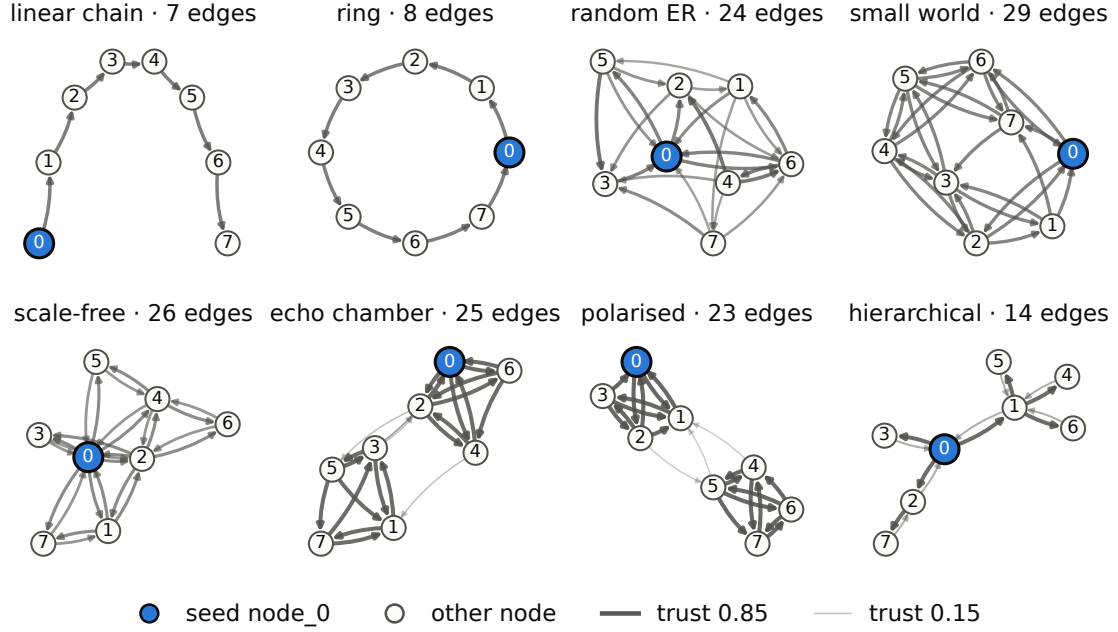


Figure 4.2: Realised eight-node graphs, one per network condition, read from the archived topology files of the T2c H `climate_scientist` configurations. The chain, ring, and tree are deterministic; the echo-chamber and polarised graphs are fixed by the seeded generator and identical across configurations; the Erdős–Rényi, small-world, and scale-free generators produce a fresh realisation per configuration. Directed edge counts: linear chain 7, ring 8, Erdős–Rényi 24, small world 29, scale-free 26, echo chamber 25, polarised 23, hierarchical 14. Edge width and opacity scale with initial trust, and the seed `node_0`, which is also the root of the hierarchical tree, is filled.

Each network condition bundles a graph with an action policy and a threshold, and the bundle differs between the two path-like conditions and the other six (Table 4.2): the chain and ring use a rewrite-heavy policy under a lower trust threshold than the other six conditions. The three clustered conditions additionally use cluster assignment (Section 4.3), and the Erdős–Rényi builder ignores the seeded random generator and the minimum seed out-degree that its configuration records. A difference between the ring and the scale-free condition therefore reflects a different rewrite rate, threshold, and graph at once, so the thesis speaks of network conditions rather than topologies, and Chapter 5 reports condition orderings as bundled-protocol contrasts, never as differences attributable to the graph generator alone.

The chain and ring expose the same processing path within the window. Because delivery is synchronous and `node_0` is seeded before tick 1, node k first processes at tick $k + 1$; the ring’s closing edge would return the message to `node_0` for tick 9. In all 36 ring runs, `node_0` recorded no event beyond its seed events. Their score differences therefore reflect different action, generation, and audit realisations within eight ticks, without observing the cycle’s return.

4.5 Seeding and propagation protocol

Propagation is synchronous, bounded to eight ticks, and identical across all 288 configurations except for the bundled parameters just described. Messages produced during tick t are delivered only after all nodes have finished that tick and become available at tick $t + 1$. Every node is always active, and its inbox holds at most four messages per article. The run stops early when a tick produces no outgoing message and otherwise after tick eight, so archived events carry input hop counts from zero to seven.

The trust filter uses the receiving node’s own relation to the sender, not the trust on the edge that carried the message; that relation exists only when a reverse edge exists, and otherwise the engine substitutes 0.50, which exceeds both thresholds. Once a message passes the hop and trust filters, the node samples an action with the bundled probabilities, and persona prompts affect the text only when a rewrite is sampled. Persona similarity can nevertheless affect whether an admitted message passes the trust filter. At each action draw, the rewrite probabilities are 0.85 for the chain and ring and 0.50 for the other six conditions. Realised rewrite totals also depend on drops, reach, and branching.

Trust evolution was enabled but could not affect actions in the same run. The configurations set `relationEvolution` to true with step 0.05, and the auditor moves the receiving node’s stored trust in the sender by that step according to whether the audited MI is at most 3 or above 3. Because propagation of all six articles completes before auditing begins, these updates enter the final node state only after every action in the run has been sampled. Belief updating, frame analysis, strategic agents, network evolution, opinion dynamics, institutional trust, competing article groups, and interventions were disabled in every Phase 2 configuration.

4.5.1 Inbox admission rule

The inbox cap is a deterministic exposure policy, not random thinning. Within a tick the engine processes nodes in stored order (`node_0`, `node_1`, and so on), appends their outgoing messages to one delivery queue, and applies the queue in order once all nodes have finished. Because a recipient admits at most four messages per article and the engine checks the cap at delivery, it admits the first four deliveries in sender order and discards the remainder without any event, counter, or log line. Earlier senders therefore have priority when their deliveries compete for a full inbox.

The admission policy can be reconstructed from the event record because every admitted message yields one event at the recipient. Proposed deliveries at tick $t + 1$ are the forward and reinterpret events at tick t from nodes with an edge to that recipient. Admitted deliveries are the events the recipient recorded at tick $t + 1$. The audit compares their count with the smaller of four and the proposed count, then checks sender order whenever the cap binds. Section 5.4 reports the realised rejection counts and their association with persona placement; Table D.3 gives the condition and seat breakdowns.

4.6 Agent and auditor instruments

Both the rewriting agents and the auditor call the same model alias, `gpt-4o-mini`, through a client without retries, using temperature 0.70, a 700-token completion cap, a 120-second timeout, no API seed, and no response schema. The standard runner executed two configurations concurrently, and response identifiers, model fingerprints, and raw auditor replies were not archived. The auditor receives the propagated text, the article’s five questions, and the expected answers and returns one score per item; its scores are judgements by a language model rather than human annotations, with the known dependence of such judgements on the judge (Zheng et al., 2023).

The two auditor regimes differ in the item scale and hence in the arithmetic of the index. In the discrete instrument the auditor returns $s_j \in \{-1, 0, +1\}$ for item j , meaning contradiction, missing information, or a correct statement, and with $m = 5$ items the index counts the items that are not correct:

$$\text{MI}_d = n_{\text{missing}} + n_{\text{incorrect}} = \sum_{j=1}^m \mathbb{I}(s_j \leq 0), \quad \text{MI}_d \in \{0, 1, \dots, 5\}. \quad (4.1)$$

In the continuous instrument the auditor returns $q_j \in [0, 1]$, where 1 denotes a fully correct statement, 0.5 missing or neutral information, 0 contradiction, and intermediate values partial accuracy; the index is the complement of the mean item score scaled to the same nominal range:

$$\bar{q} = \frac{1}{m} \sum_{j=1}^m q_j, \quad \text{MI}_c = m(1 - \bar{q}), \quad \text{MI}_c \in [0, 5]. \quad (4.2)$$

The continuous record also derives correct, missing, and incorrect counts by cutting q_j at 0.67 and 0.33, but those rates do not enter the analysis tables.

The two scales are not commensurate, because a missing item adds one point under the discrete rule but half a point under the continuous rule, whereas a contradicted item adds one point under both. A text that omits all five items therefore scores 5 on the discrete scale and 2.5 on the continuous scale, and a text that contradicts all five scores 5 on both. With five items, omissions alone cannot cross the inherited cut at $MI > 3$ under the continuous rule. Contradictions and omissions combine, however: two contradicted and three missing items give $MI_c = 2 + 3 \times 0.5 = 3.5$, and four contradictions are required only when every remaining item is fully correct. The cut is therefore not threshold-equivalent across instruments, and every quantity derived from it, including k^* (Section 4.10), is instrument-conditional. The discrete rubric offers only three item categories, which limits resolution (MacCallum et al., 2002). It is not a post-hoc coarsening of the continuous scores: the two prompts elicit separate judgements and assign different penalties to missing items.

The dual regime issues both auditor calls in parallel on the same text and reports the discrete score as the headline. The event’s top-level index is MI_d , and a sidecar stores both records, an absolute gap, and a within-event item-score correlation (the engine field `dual.agreement`):

$$g_e = |MI_{d,e} - MI_{c,e}|, \quad (4.3)$$

$$a_e = \text{corr}\left(\frac{s_{1,e}+1}{2}, \dots, \frac{s_{m,e}+1}{2}; q_{1,e}, \dots, q_{m,e}\right), \quad (4.4)$$

where the discrete items are mapped from $\{-1, 0, +1\}$ to $\{0, 0.5, 1\}$ before Pearson’s coefficient is computed. The gap is a scorer-disagreement diagnostic, not a third index. Its cell-level summary averages absolute event-level differences within each cell; the campaign summary then weights live cells equally unless event pooling is stated. The correlation measures association rather than exact category agreement and is undefined whenever either item vector is constant. It can therefore exclude perfectly agreeing all-correct vectors. Section 5.2 reports the eligible event and cell counts alongside the observed correlations.

The comparison separates three components: the algebraic difference in the missing-item penalty; disagreement between stochastic judge calls using different rubrics on the same text; and differences between separately executed campaigns. The sidecar combines the first two components while holding generated texts fixed. A T2c–T2d headline contrast additionally changes realised trajectories and scoring coverage, so it cannot identify the contribution of the scoring rule. Chapter 5 therefore presents the sidecar evidence first and keeps the campaign headlines separate.

4.6.1 Unchanged-source control

An event whose output is byte-identical to the source article cannot have lost or distorted a source claim. A valid claim-recoverability score should therefore be zero, making such events a negative control. The control includes all scored source-identical events in the retained grid, including those in dead cells: 5,345 events in T2c and 4,237 in T2d. Each T2d event receives both a discrete assessment and a continuous sidecar assessment. Of the

9,582 events, 5,159 are forwards and 4,423 are reinterpretations, including failed-rewrite fallbacks (Section 4.11); they occur at ticks 1–8. Checks using whitespace normalisation or sequence similarity of at least 0.98 found no additional near matches.

The control is stratified by article, instrument, and item, and its counts are repeated audits within single runs, not independent replicates. Unchanged-source events form 16.0, 2.8, 9.0, and 8.5 per cent of the scored events in the live cells of T2c H, T2c He, T2d H, and T2d He, and the control reports the observed denominators for each article and item. Section 5.2 reports its outcome first among the RQ1 results (Table A.3 and Figure 5.1), together with a sensitivity that recodes verified-identical events to $MI = 0$; the recoding measures the contribution of these observed control scores to each headline mean. It neither bounds errors on altered texts nor calibrates the instrument.

4.7 Campaign design and replication

The main campaign is a full factorial of network condition, persona condition, instrument, and article, and its unit is the configuration–article cell. Table 5.1 lists the four slices. Eight network conditions crossed with 12 homogeneous personas and two instruments give 192 configurations, and crossed with six mixes and two instruments give 96 more; each configuration seeds six articles, so

$$\underbrace{8 \times 12 \times 2 \times 6}_{\text{homogeneous}} + \underbrace{8 \times 6 \times 2 \times 6}_{\text{heterogeneous}} = 1,728 \text{ cells} \quad (4.5)$$

across 288 configurations. The D-net stress test adds four configurations and eight cells that are analysed separately (Section 4.13).

The design retains one run per configuration and therefore provides no empirical distribution across independent replicates of that configuration. Its six article cells share one realised graph, one seating, and one action policy, with fresh action draws for messages that pass the processing filters. They are repeated observations from a single simulation; the eight network conditions are fixed design strata. The campaign supports descriptive comparisons of executed cells but cannot estimate seed-to-seed variability (Chapter 6).

Reruns exist for 77 of the 288 experiment names. A manifest fixes the retained directory originally selected as the newest complete directory with at least one successful call. It replaces that timestamp rule because checkout changes timestamps; reapplying the rule would select different directories for nine configurations (Section C.3). All thesis tables use the reconciled data regenerated from the manifested node histories. They differ from the historical table in the two cells documented in Section C.3.

Because most of the randomness was unseeded, the archived record is traceable but not replayable. Every configuration records `graphRandomSeed` 42, but only the echo-chamber and polarised builders consume it; the Erdős–Rényi, small-world, and scale-free builders and the action sampler call JavaScript’s unseeded `Math.random`, and the language model’s outputs vary between calls. The archive records what was executed, and a fresh run

of the same configuration is a protocol replication rather than a bitwise reproduction (Chapter A).

A cost constraint fixed the eight-tick horizon, which limits the analysis to early propagation (Section 4.14). The prior study used 30-hop linear chains (Maurya et al., 2025). Here, dense and cyclic graphs can generate up to 32 events per article per tick; each reinterpretation requires a paid rewrite, and each scored event requires one auditor call under the continuous instrument or two under the dual instrument. The eight-node, eight-tick design trades trajectory length for coverage of configurations. It does not establish convergence or equivalence to 30 hops. Section 5.4.4 examines event activity and score changes at the observation boundary.

Separate directory trees isolate the campaign from the pilot. The parser selects runs by experiment name within the Phase 2 run tree only, so the discrete-only pilot (Section C.1), probe runs, and D-net runs cannot enter the 1,728-cell grid.

4.8 Raw records and derived cells

The reported cell estimates derive from the retained event histories. Two histories end during auditing; their reproducible retained-history estimates replace completed-run estimates whose full event records are unavailable (Section C.3). Raw auditor replies were not archived (Section 4.6).

Each retained directory contains metadata, the merged configuration, usage counters, the realised graph and initial trusts, and one node file per seat. Node files record persona identifiers, relations, action counters, and event histories; checkpoint and per-article result files provide additional summaries. The parser reads event histories rather than aggregate result files. Each event records the tick, article, sender, hop count, action, input and output text, audited index, reason code, and provenance chain. A failed audit leaves the index null; a successful audit also stores the full score object.

The cell headline is the event-weighted mean of the scored events, and a node-weighted companion is stored beside it. Let E_a^* be the events of article a in a configuration whose top-level index is not null. The parser reports

$$\overline{\text{MI}}_a = \frac{1}{|E_a^*|} \sum_{e \in E_a^*} \text{MI}_e, \quad (4.6)$$

which the tables call **meanMI** and this thesis calls the live-cell mean MI. The node MPR of node v is the mean index of that node's scored events for the article; the companion field **meanNodeMPR**, written $\overline{\text{MPR}}_a$ and called the mean node MPR, averages it over the nodes V_a^* with at least one scored event:

$$\text{MPR}_{v,a} = \frac{1}{|E_{v,a}^*|} \sum_{e \in E_{v,a}^*} \text{MI}_e, \quad \overline{\text{MPR}}_a = \frac{1}{|V_a^*|} \sum_{v \in V_a^*} \text{MPR}_{v,a}. \quad (4.7)$$

Chapter 5 reports the live-cell mean as the headline and the mean node MPR as a companion, never substituting one for the other; the parser also stores the maximum event index, the number of scored events, and the k^* fields of Section 4.10.

4.9 Scoring eligibility and analysis population

A cell with at most one scored event is classified as dead and is hatched rather than deleted or set to zero. Formally, $\text{dead} \iff n_{\text{scored}} \leq 1$; rows with positive model usage are labelled `hatched_dead_after_llm`, remain in the cell table and the dead-cell ledger, are drawn hatched in figures, and are excluded from every live-cell mean. A dead cell is not an observation of $\text{MI} = 0$ and not evidence that the seated composition prevented propagation, because the same status arises when few messages were generated and when auditor requests failed (Section 4.11).

Of the 1,728 cells, 292 (16.9 per cent) had at most one scored event: 96 (T2c H), 98 (T2d H), 50 (T2c He), and 48 (T2d He). Of these, 232 have no scored event and 60 have exactly one. The corresponding live populations are 480, 478, 238, and 240 cells. Propagation means, medians, paired contrasts, sign counts, and k^* rates condition on these live sets; full-grid and dead-cell counts are labelled separately. The unchanged-source control instead includes all scored source-identical events, including nine T2c and 16 T2d events in dead cells (Section 4.6.1).

Live-cell means describe cells that survived both propagation and auditing. Their dead-cell rates must accompany them because a low conditional score alone does not establish successful preservation across the full grid. A successor design should therefore report scorable-cell survival and conditional severity separately (Section 6.6).

4.10 Within-window k^*

The threshold statistic k^* marks the start of the terminal above-threshold suffix of a cell's observed scored ticks: it is the earliest scored tick from which the network mean index exceeds 3 at every later scored tick within the window. For each cell the parser forms the tick series $\overline{\text{MI}}_t = |E_t^*|^{-1} \sum_{e \in E_t^*} \text{MI}_e$ over scored events and defines

$$k^* = \min\{t : \overline{\text{MI}}_t > 3 \text{ and } \overline{\text{MI}}_u > 3 \text{ for every later scored tick } u\}. \quad (4.8)$$

A cell with no qualifying suffix has k^* undefined, stored as null rather than zero. A defined k^* exists if and only if mean MI at the last scored tick exceeds 3; its occurrence rate is the share of live cells meeting that endpoint condition. The value of k^* identifies the starting tick of the terminal above-threshold suffix, which may contain only the final scored tick (Table D.2). That tick need not be tick eight: propagation may stop earlier, or later events may receive no score. Thus, k^* does not measure persistence beyond the window. Each campaign computes it from its own headline series, and their rates are reported separately.

The statistic is a simulator threshold under the inherited cut, not the onset time of an empirical firestorm. It inherits the arbitrariness of the cut at 3 (Section 3.2) and its non-equivalence across instruments (Section 4.6); it is defined on ticks rather than clock time, and it is censored at eight ticks. Chapter 5 therefore attaches no timing interpretation to it.

4.11 API failures and scoring coverage

The campaign recorded 203,928 successful model responses and 64,270 HTTP-error responses across the 288 retained runs. The error figure sums the per-run `llmUsage.errors` counters, each of which increments once when a request receives a status of at least 400 without retry, so the figure is neither a retry count nor a duplicate over reruns. It is not a request ledger: request identifiers were not retained, the dual auditor’s two concurrent requests can hide a sibling rejection from the event-level handler, and 7,133 counter increments have no matching caught log line. Among the caught lines, 57,137 report HTTP 429 and one a transport error, components that the log records separately and that need not sum to the counter total. Errors touched 121 of the 288 runs, and the engine marked no retained run as aborted or failed.

Three error-handling paths affect which outputs enter the analysis. When a rewrite request fails, the node writes the received text back as its output, keeps the label reinterpret, and propagates the unchanged text, whose later audit enters the cell mean like any other; the final states contain 13,585 such identity events, close to the 13,494 caught rewrite failures, although 211 occur in error-free runs and the two sets cannot be matched one to one. When an auditor request fails, the event’s index remains null and the parser excludes it from the cell mean, the node MPR, and k^* ; of 129,664 audit-eligible events, 93,470 were scored and 36,194 remained null. When an auditor reply cannot be parsed, the engine fails open and scores every item as correct; the retained logs contain no parse-fallback warning, so the observed count of this path is zero.

The failures were clustered in time and by condition. Runs that started in the 02:00 UTC hour recorded no errors, runs that started at 03:00 recorded 6,853, and runs that started at 04:00 recorded 57,417 of the 64,270, an error share of 44.7 per cent of attempts in that cohort. By slice the shares were 28.1 per cent (T2c H), 18.3 per cent (T2c He), 21.5 per cent (T2d H), and 27.0 per cent (T2d He). The linear chain and ring conditions recorded no HTTP error in any slice, whereas the hierarchical condition in T2c H lost 59.7 per cent of its attempts. Because unscored events reduce n_{scored} , the dead-cell counts of Section 4.9 conflate sparse propagation with unavailable audits, and dead-cell rates by condition partly reflect the distribution of API errors.

Two restrictions assess how scoring coverage changes the descriptive comparisons. The first keeps runs whose HTTP-error counter is zero and recomputes the live-cell means and He–H differences. The second keeps cells in which every audit-eligible event received a score. These restrictions select on condition and execution time; neither isolates an API effect. Section 5.4.4 and Chapter D report their populations and the direction of the principal findings.

The sensitivity analyses preserve the retained-run manifest rather than substitute reruns. Only 12 experiment names have a complete alternative without HTTP errors, and only five of these would replace an affected retained run.

Table 4.3: Comparison families. Each row names the compared slices, the dimensions held and varied, and the rule that bounds what the contrast may be read to show.

ID	Compared slices	Held and varied	Interpretation rule
C1	T2c H vs T2c He; T2d H vs T2d He	network condition and instrument held; composition varied	same-regime composition contrast, estimated separately for each instrument; a bundled prevalence, placement, and assignment contrast
C2	T2c H vs T2d H	network condition and homogeneous arm held; instrument varied	instrument comparison across separate campaigns; not a common-scale treatment effect
C3	T2c He vs T2d He	network condition and heterogeneous arm held; instrument varied	as C2; the dual sidecar supplies the same-event comparison and is reported separately
C4	T2c H vs T2d He; T2d H vs T2c He	network condition held; composition and instrument both varied	two labelled, confounded contrasts shown in both directions; no pooled C4 headline
C5	live cells concatenated within a labelled slice or composition subset; D-net separate	table union across the eight generated conditions	descriptive pooled summaries with cell-equal and condition-equal weights; no physical super-graph

4.12 Comparison families

Five comparison families organise the descriptive analysis; they were defined after the campaign had been parsed and before the results chapters were written (Section 4.14), and Table 4.3 names what each holds and varies. In every family a same-condition comparison aligns summaries by one of the eight network-condition labels and does not imply that the T2c and T2d runs share event histories (Section 4.6).

The families differ in how they weight and pair cells. C1 computes H and He means within each network condition and instrument and reports both a cell-pooled difference, which weights each live cell equally, and a condition-equal difference, which weights each of the eight condition means by one eighth. C2 and C3 additionally pair live cells joined on network condition, persona or mix, and article; a pair enters only when both cells are live, so a dead cell on either side is never replaced by zero. C4 crosses composition and instrument deliberately, and its two directions are shown separately because their average would be a third, uninterpretable quantity.

C5 pools live cells across the eight network conditions within a labelled instrument and composition subset. It averages their recorded outcomes without joining their graphs or rerunning propagation. D-net cells are analysed separately.

4.13 D-net construction and structural diagnostics

The D-net is a constructed hashtag co-occurrence graph, not a retweet cascade, and it carries no empirical propagation outcome. Two data pathways were attempted, and

neither produced a user-level dataset. First, hydration of the tweet identifiers deposited by Debnath (2023a) did not run: no platform credential was available, the Mendeley files interface returned an error, and the identifiers in the related OSF deposit are stored in a lossy floating-point notation from which none of 1,786 sampled tokens could be recovered. Second, the author’s script streamed the first 5,000 records of `dataset.csv`, version 1 in the OSF project *Geoengineering on Twitter* (Debnath, 2022b), once and parsed them in memory; it extracted hashtags from the text column with the regular expression of the published analysis code (Debnath, 2023b) and discarded the text, user, and profile fields without writing them to disk. Only aggregates were retained and published, namely per-hashtag and per-hashtag-pair counts (564 distinct hashtags; 987 records with at least one hashtag; 65 documented tag pairs observed) and a diagnostic record of the identifier format (Section 6.7).

The graph itself was built from the hashtag clusters, co-occurrence pairs, and cluster statistics documented in the article and its analysis code (Debnath et al., 2023; Debnath, 2023b): each of its 63 nodes is a hashtag-defined role, not an observed account; its 228 directed edges (171 unique undirected pairs) connect roles whose hashtags co-occur or share a cluster; the OSF counts annotate 48 of these edges and added or removed none; and the seed is `user_chemtrails_hub`. The object has no tweet clock and no empirical MI or MPR, so the thesis calls it a stress test and claims neither empirical validation nor platform fidelity for it.

Four configurations were executed on the same edge set. The homogeneous pair seats `conspiracy_believer` at every node; the mixed pair seats each node according to its hashtag cluster, giving 28 conspiracy, 19 climate-action, 13 environmental-concern, and 3 expert seats, of which 56 carry the exact persona identifier that the cluster rule intended and seven a coarser identifier of the same family. Both assignments were run under the continuous and the dual instrument with the six-condition action policy (0.35, 0.50, 0.15; $\tau = 0.15$), eight ticks, a hop cap of eight, and two seed articles, `scopex_2017` and `chemtrails_gates_2018_2021`, which gives eight cells.

The construction has global transitivity 0.31, mean local clustering 0.52, and 33 unique neighbours at the seed. Family homophily is 0.84 under the mixed assignment and 1.00 under homogeneous assignment by construction. These describe the input graph rather than emergent simulation outcomes.

The structural comparison uses depth, breadth, and structural virality (Goel et al., 2016). Its reference is a breadth-first traversal of this same input graph, not an independent observed cascade. A corrected extractor builds each simulated first-receipt tree from the recorded sender, rooted at the seed. Chapter B documents the original parent-extraction defect, the corrected values, and the limits of comparing propagation with its own input wiring. A close match cannot establish platform fidelity.

4.14 Proposal and executed scope

The April 2026 proposal envisaged a more direct connection to observed geoengineering discourse and firestorm mechanisms. The executed study supplies a synthetic measure-

ment audit. The principal departures are the replacement of a hydrated user corpus by curated stimuli and theme-informed personas, the use of a constructed hashtag graph for the larger-network exercise, the eight-tick budget, and the absence of planned independent replication or a full firestorm outcome model. Section C.1 records these changes and their consequences without asserting supervisory approval.

4.15 Statistical stance and artefact register

The analysis reports descriptive counts, live-cell means, paired medians, sign counts, threshold rates, and Spearman condition-rank correlations. Each describes the retained campaign. Fixed strata, nested article cells, and one run per configuration preclude repeat-run inference; exploratory test statistics are not used as evidence. Missing heat-map observations remain blank.

The companion archive supplies the fixed run manifest, reconciliation code, derived tables, and manuscript sources. Sections C.3 and C.4 document regeneration from retained histories without new model calls and distinguish it from exact replay.

Chapter 5

Results

The results follow the four research questions after establishing the scored population. The main grid contains 288 configurations and 1,728 article cells; D-net is analysed separately. Same-event scoring comparisons precede composition and network contrasts because the latter depend on the instrument. All summaries follow the analysis rules in Chapter 4.

5.1 Campaign overview and integrity

Three properties condition the propagation summaries. Each configuration contributed one retained run. Its article cells share a graph, persona assignment, and action policy, with actions sampled after the processing filters; the 1,728 cells are not independent replicates (Section 4.7). The campaigns were executed separately, so headline contrasts combine scoring rules with new realisations and scoring coverage. Paired discrete and continuous scores within T2d compare identical generated texts (Section 4.6). Finally, cells with at most one scored event were excluded from propagation means and k^* rates without zero imputation. Those summaries therefore condition on propagation and successful auditing (Section 4.9).

The four campaign slices lost similar shares of cells but separated by instrument before they separated by arm (Table 5.1). Dead shares ranged only from 16.7 to 17.4 per cent, which does not establish comparable composition or score distributions among the retained cells. Live-cell means did separate: both T2d slices lay above both T2c slices, and the mean node MPR tracked the live-cell mean to within 0.01 in all four pooled slices. The k^* rates did not follow the means: the He arm had the lowest rate under the continuous and the highest under the dual-discrete instrument, a reversal to which Section 5.2 shows the scoring rule contributes.

The 292 hatched-dead cells, 232 with no scored event and 60 with one, were unevenly distributed across network conditions (Table 5.6). Across conditions, the dead share ranged from 5.6 per cent for the linear chain in T2d H to 30.6 per cent for the polarised condition in T2c He, and the hierarchical condition lost 27.8 per cent in both T2c arms. The hierarchical and polarised conditions, which have the highest continuous He live-cell means (Section 5.4), thus rested on the smallest survivor populations, so the reported means must be read with their live-cell counts and survival summaries.

Scoring coverage, not only sparse propagation, shaped which cells survived. The API audit (Section 4.11) records 64,270 HTTP-error responses, including 57,417 on runs

Table 5.1: Campaign overview by slice: cells, live cells, hatched-dead share, live-cell mean MI, mean node MPR, and share of live cells with a within-window k^* . Dead shares are similar; means and k^* rates separate by instrument.

Slice	Cells	Live	Dead (%)	Live-cell mean MI	Mean node MPR	k^* rate (%)
T2c H	576	480	16.7	0.86	0.86	14.2
T2c He	288	238	17.4	1.02	1.01	7.6
T2d H	576	478	17.0	1.68	1.68	20.1
T2d He	288	240	16.7	2.08	2.07	32.1
All	1,728	1,436	16.9	—	—	—

started in the 04:00 UTC hour and none in the 02:00 cohort. Runs were scheduled by slice and condition; within T2c H, the error burden ranged from zero for the chain and ring to 59.7 per cent for the hierarchical condition. The dead shares in Table 5.6 therefore mix sparse propagation with unavailable audits. Section 5.4 applies the zero-HTTP-error check to the arm contrast, and Section 5.4.4 assesses the principal findings under restricted scoring populations.

5.2 RQ1: sensitivity to the auditor regime

RQ1 asks how sensitive claim-recoverability levels, within-window k^* occurrence, and network-condition orderings are to the continuous and dual-discrete auditor regimes. The analysis keeps three components apart: (a) the algebraic scoring-rule difference, under which a missing item adds 1 discretely and 0.5 continuously while a contradicted item adds 1 under both (Section 4.6); (b) same-text auditor disagreement, two stochastic judge calls with different rubrics on identical texts, observed through the continuous sidecar on successfully scored dual events; and (c) separate-campaign outcome differences between the T2c and T2d realisations. The unchanged-source control (Section 4.6.1) is reported before any of them.

5.2.1 Unchanged-source control

The unchanged-source control, using all scored source-identical events including those in dead cells, separated the two regimes sharply (Table A.3 and Figure 5.1). An event whose output text is byte-identical to the curated article cannot have lost or distorted a source claim, so a valid claim-recoverability score should be zero on it (Section 4.6.1). Both continuous readings were near zero: the headline auditor returned a nonzero MI on 0.7 per cent of 5,345 unchanged T2c audits and the sidecar on 1.0 per cent of 4,237 unchanged T2d audits, every deviation confined to three expected-‘No’ items, Paris item 2 and SAI items 4 and 5. The discrete headline auditor returned a nonzero MI on 38.4 per cent of the same 4,237 audits, with a mean MI of 0.51 on text identical to the source, concentrated on the polar-bear, Paris, and SAI articles and absent from SCoPEX and chemtrails.

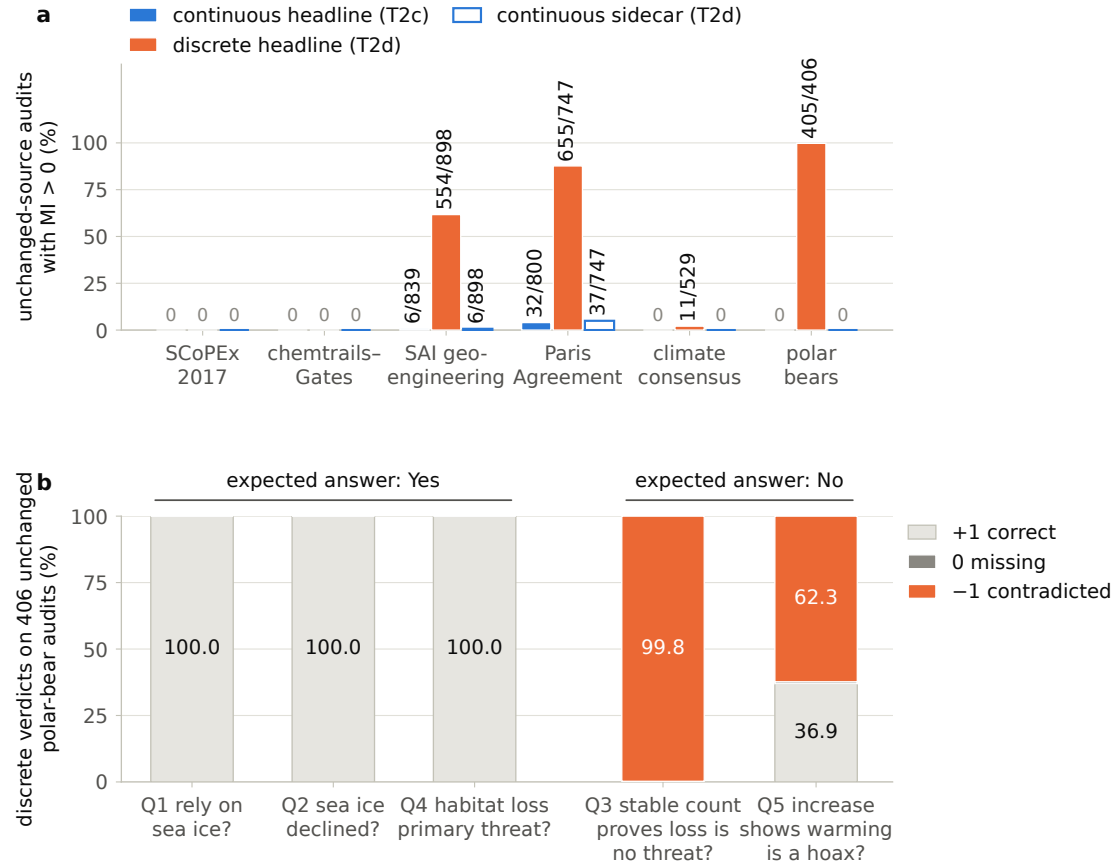


Figure 5.1: Unchanged-source control across all retained cells, including dead cells: (a) share of unchanged-source audits with nonzero MI by article and instrument (bar labels: nonzero over total audits); (b) item-level discrete verdicts on the 406 unchanged polar-bear audits by expected answer (Yes: items 1, 2, 4; No: items 3, 5). Discrete departures from zero occur only on expected-‘No’ items; continuous departures are much rarer.

The failure was specific to a subset of the expected-‘No’ items. All 2,157 non-correct discrete verdicts on unchanged text fell on those items; none of the 11,813 expected-‘Yes’ verdicts failed. Six of the 13 ‘No’ items also never failed. Negative polarity was therefore necessary but not sufficient for failure in this control.

The sidecar usually assigned a different verdict to the same text and item. Of 2,106 discrete -1 verdicts on unchanged text, 2,073 received a continuous score of 1.0. One example occurs in run `T2d_H_echo_chamber_climate_action_advocate: node_0` forwarded the polar-bear article unchanged at tick 1 (history entry 90). Although the source explicitly gives the expected ‘No’ to item 3, the discrete verdicts $[1, 1, -1, 1, 1]$ produced $MI = 1$; the sidecar returned $[1, 1, 1, 1, 1]$ and $MI = 0$ in a separate judge call. This pattern is consistent with polarity confusion when the source refutes a proposition in other words. No controlled within-rubric rescoring experiment tested that mechanism.

Unchanged-source events constituted 16.0, 2.8, 9.0, and 8.5 per cent of scored events in live T2c H, T2c He, T2d H, and T2d He cells, respectively. Recoding these events to zero left the T2c slice means unchanged at two decimals and reduced the T2d H and He means from 1.68 to 1.63 and from 2.08 to 2.02. This is a sensitivity scenario, not a calibrated correction.

The recoding preserved family and mix orderings but changed some condition rankings. Polarised H fell from rank 3 to 7 among the eight T2d conditions, while echo chamber and small world exchanged ranks in T2d He. Substantive discrete-regime rankings therefore remain conditional on the observed scoring defect. The control does not validate either instrument on altered texts.

5.2.2 Same-text disagreement in the dual sidecar

The discrete headline usually exceeded the continuous sidecar when both assessed the same generated texts (Figure 5.2). The difference between cell means was positive in 431 of 478 H cells (90.2 per cent) and 224 of 240 He cells (93.3 per cent). Weighting cells equally, the mean signed difference was 0.84 in H and 1.05 in He. Averaging absolute event-level differences within each cell, then weighting cells equally, gave gaps of 0.88 and 1.11; pooling events instead gave 0.87 and 1.10 (Table 5.2). These gaps concern the scoring procedures because the compared texts are identical.

The larger omission penalty accounts algebraically for part of the signed difference. Recoding each event’s discrete verdicts onto the continuous anchors gives an omission component of 0.32 in H and 0.61 in He, with residuals of 0.52 and 0.44. Those residuals combine rubric-dependent and stochastic judge variation: the two assessments used separate calls at temperature 0.70. Altered texts were not systematically rescored within each rubric, so this audit cannot separate those sources of variation.

The decomposition also varies by condition. The ring and chain had the largest He signed gaps (1.27 and 1.23), mostly attributable algebraically to omission penalties (0.79 and 0.76). The three cluster-assigned He conditions had gaps of 0.84 to 0.91 and omission components of 0.40 to 0.51. The eight H cells and one He cell with negative signed differences further show that the two judges sometimes assigned different item states to identical texts.

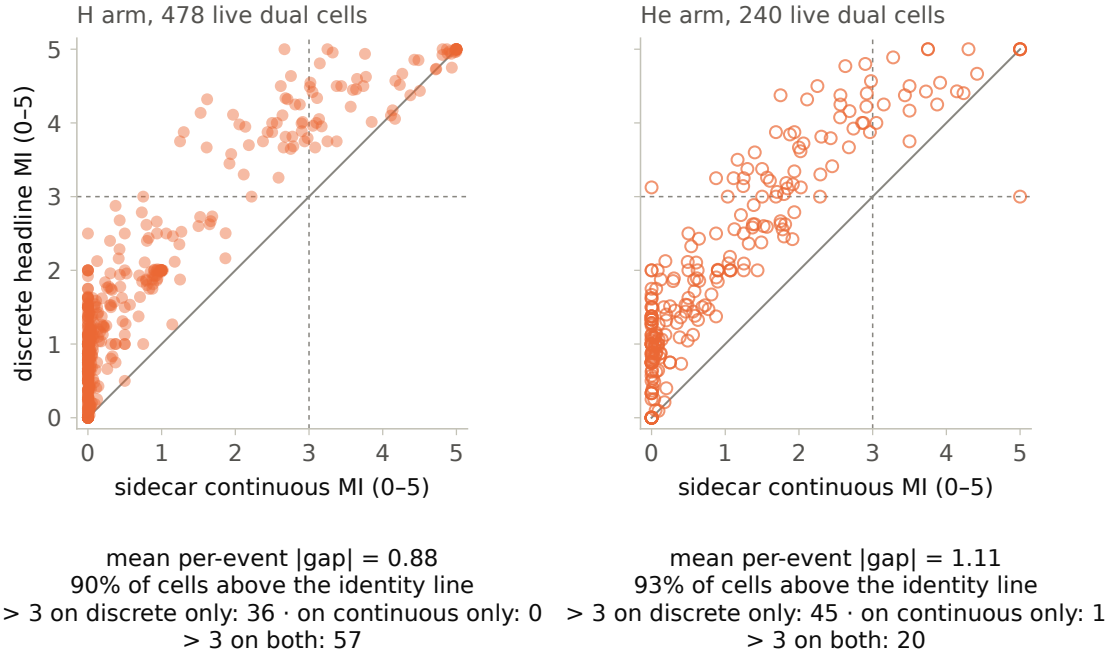


Figure 5.2: Cell-mean discrete headline (vertical axis) against cell-mean continuous sidecar (horizontal axis) for every live dual cell (478 H, 240 He). Most points lie above the identity line, increasingly far at higher headline levels.

Table 5.2: Signed same-event dual gap and its decomposition, live T2d cells: $\text{MI}_d - \text{MI}_c = 0.5 n_0(s) + [c^*(s) - \text{MI}_c]$, with $n_0(s)$ the number of items the discrete auditor scored 0 (missing) and $c^*(s) = n_{-1}(s) + 0.5 n_0(s)$ its verdicts recoded onto the continuous anchors; equal-cell rows weight cells equally, equal-event rows pool events. The omission penalty is the smaller part in H and the larger in He.

Arm	Weighting	Cells	Events	Total	Signed gap $\text{MI}_d - \text{MI}_c$		Absolute gap
					Omission penalty $0.5 n_0$	Residual $c^*(s) - \text{MI}_c$	
H	equal cell	478	34,388	0.837	0.316	0.521	0.880
H	equal event	478	34,388	0.819	0.329	0.490	0.875
He	equal cell	240	13,470	1.050	0.610	0.440	1.115
He	equal event	240	13,470	1.048	0.622	0.427	1.099

The item-score correlation covers only a selected subset of scored events. Pearson’s coefficient between the five recoded discrete verdicts and continuous item scores was defined for 8,511 of 34,388 H events (24.8 per cent) and 4,345 of 13,470 He events (32.3 per cent). At least one event qualified in 270 of 478 H cells and 171 of 240 He cells. Averaging the eligible event coefficients within each cell, then weighting those cells equally, gave 0.65 and 0.62; pooling eligible events gave 0.67 in both arms. Constant vectors were excluded, including 11,318 H events scored all-correct by both judges.

Comparing item categories complements the correlation analysis. The continuous auditor used 0, 0.5, or 1 for 99.8 per cent of its item scores. Binning at its cut-points, the judges assigned the same class to about four fifths of item pairs; the discrete judge was harsher in most disagreements (Table A.2). The hierarchical He condition had the lowest item correlation (0.29) and the highest sidecar and headline means, whereas the absolute gap was largest for the chain and ring in He. These associations describe scorer behaviour rather than independently establishing factual drift.

The scoring procedures classified cell means differently at $MI > 3$. Among the 478 live H cells of T2d, 36 exceeded the cut only under the discrete cell mean, none only under the sidecar mean, 57 under both, and 385 under neither. Among the 240 live He cells, the corresponding counts were 45, 1, 20, and 174. These classifications concern whole-cell averages of the same scored events.

Applying the cut to each cell’s final scored-tick mean gives the within-window k^* classifications. In H, 96 cells had a discrete k^* and 61 had a sidecar k^* , with all 61 included in the discrete set. In He, 77 and 22 cells qualified, with 19 in both sets (Table D.2). The discrete procedure therefore classified more cells above the endpoint threshold even when the generated texts were held fixed.

5.2.3 Campaign-level contrasts C2 and C3

Across campaigns, the dual-discrete headline lay above the continuous headline on every network condition in both arms (Table 5.6). The condition-level difference averaged 0.82 in the homogeneous arm (C2) and 1.05 in the heterogeneous arm (C3); the differences ranged from 0.70 (chain) to 0.95 (echo chamber) in H and from 0.47 (polarised) to 1.38 (random Erdős–Rényi, ER) in He. The two series are not parallel, the continuous He headline rising on the three cluster-assigned conditions while the dual-discrete headline does not. These gaps match the same-event signed gaps of 0.84 and 1.05 in size, which is consistent with a large scoring contribution, but the campaign contrast does not separate scoring specification from fresh trajectories and scoring coverage.

Pairing cells across campaigns by condition, persona or mix, and article gave the same directional pattern at cell level. Most of the 398 H and 198 He cells live in both campaigns scored higher under T2d, with median paired differences of 0.81 and 0.98. In H, 80 cells were dead only in T2c, 82 only in T2d, and 16 in both campaigns; the corresponding He counts were 42, 40, and 8. Similar losses on each side do not establish that the paired subset is free of selection, because the lost cells may differ by article, persona, or condition.

5.2.4 Condition orderings on the same texts and across campaigns

Higher discrete levels did not imply the same ordering of network conditions. Across the separate campaigns, the Spearman correlation between the eight continuous and dual-discrete condition means was -0.05 in H and 0.17 in He. Within the dual campaign, ranking the same texts by headline and sidecar with equal cell weights gave 0.62 in H and 0.12 in He. Thus, H rankings agreed more closely on the same texts than across campaigns, whereas He rankings differed markedly in both comparisons.

The He ranking changes are consistent with the condition-specific omission components. The sidecar placed the three cluster-assigned conditions first. The headline retained the hierarchical condition first but placed the echo chamber sixth and the polarised condition eighth, while the chain and ring rose. Their larger omission penalties help account for this movement, although residual judge differences also contribute. Across separate campaigns, only the small-world H minimum and hierarchical He maximum remained the same (Table 5.6). Any ordering must therefore name the scoring procedure and comparison population; Section 5.4.4 examines sensitivity to scoring coverage.

5.2.5 Within-window k^* as an endpoint indicator

The k^* rates reversed the arm ordering between instruments (Table 5.1). The He arm had the lowest share of live cells with a within-window k^* under the continuous instrument and the highest under the dual-discrete instrument, a swing of 24.5 percentage points against 5.9 for H. The same-event tallies above show that scoring specification contributes to this reversal; generated texts and scoring coverage may contribute as well.

The occurrence of k^* describes the endpoint of the observed window. It exists if and only if the final scored tick's mean exceeds three (Section 4.10). Table D.2 reports the available follow-up: the above-threshold sequence consisted of the final scored tick alone in 7 of 68 T2c H crossings, 3 of 18 T2c He crossings, 5 of 96 T2d H crossings, and 14 of 77 T2d He crossings. Most crossings had further scored ticks after k^* , but none establishes persistence beyond eight ticks. The appendix table retains the full follow-up distribution; terminal-only crossings were proportionally most frequent in He, where the instrument difference in occurrence was greatest.

The deliberately confounded cross-regime contrast (C4) changes arm and instrument together. Its condition-equal differences were $+1.22$ for T2d He minus T2c H and -0.65 for T2c He minus T2d H. It is retained as a diagnostic of confounding, not interpreted as a composition effect.

5.3 RQ2: persona composition

RQ2 asks, within each auditor regime, how live-cell mean MI and within-window k^* occurrence vary across named persona families and heterogeneous compositions, and how these patterns compare with the pooled H–He contrast. The arm labels describe an assignment procedure (Section 4.3): H placed one persona on all eight nodes and so pooled twelve personas from three families; He placed a declared mix of eight persona

identifiers and so pooled six mixes with zero to four conspiracy seats. The section compares families, mixes, and articles.

5.3.1 Persona families

The conspiracy family scored highest in the homogeneous arm under both instruments (Table 5.3). Its live-cell mean MI was 2.22 under continuous scoring and 3.17 under dual-discrete scoring, about two points above either other family. The corresponding k^* rates were 40.5 and 59.7 per cent, compared with one climate-action crossing (0.8 per cent, dual-discrete) and none in the science and environment family. Every persona scored higher in the dual-discrete campaign: its eight non-conspiracy persona means ranged from 0.67 to 1.31, whereas the four conspiracy means ranged from 2.68 to 3.36. Under each instrument, the gap between these groups exceeded the spread within any family.

Family means explain why the pooled H–He contrast cannot identify a diversity effect. The pooled H means (0.86 continuous, 1.68 dual-discrete) fell below He (1.02 and 2.08), yet the conspiracy family alone exceeded He by more than one point under both instruments. H combines four conspiracy personas with eight personas from the other two families.

The higher He mean did not coincide with a higher overall conspiracy-seat share. He declared 11 of 48 seats (22.9 per cent) and realised 115 of 384 after cluster assignment (29.9 per cent), compared with 33.3 per cent in H. These shares differ from the shares of received, rewritten, or scored events, which depend on placement, admission order, and survival.

For example, `mix_01` had a continuous mean of 1.27 on the five sequential conditions, where it realised two conspiracy seats, and 1.84 on the three cluster-assigned conditions, where it realised four. Seat-weighted averages of the full-campaign homogeneous family means were 0.66 and 1.18, using conspiracy, climate-action, and science and environment weights of 2:3:3 and 4:2:2, respectively. These descriptive references do not match network conditions or exposure and cannot identify an interaction effect. The mix also placed a conspiracy persona at the seed, so its contrast combines composition and placement.

Survival also differed by family, and not in the same direction under both instruments. The conspiracy family lost 24 of 192 cells (12.5 per cent) under the continuous instrument against 27 of 144 (18.8 per cent) and 45 of 240 (18.8 per cent) for the climate-action and science and environment families, whereas under the dual-discrete instrument the losses were 33 (17.2 per cent), 18 (12.5 per cent), and 47 (19.6 per cent). The family means therefore rest on survivor populations that differ by about 7 percentage points, and a high family mean did not coincide with a high loss of cells, the opposite of the He-condition pattern in Section 5.4.

The family differences were already apparent near the start of rewriting (Figure 5.3). Among live conspiracy H cells whose earliest text-changing rewrite received a score, that event exceeded $MI = 3$ in 30 of 128 cells under continuous scoring (23.4 per cent), 55 of 135 under the dual-discrete headline (40.7 per cent), and 39 of the same 135 under its sidecar (28.9 per cent). These percentages use one earliest text-changing rewrite per cell.

The event-level depth analysis uses a different population, including forwarded copies

Table 5.3: Homogeneous persona families collapsed across conditions and articles: personas, live cells n , live-cell mean MI, and k^* rate per instrument. The conspiracy family lies more than a point above the others and contains all but one live cell with defined k^* .

Family	Personas	Continuous (T2c)			Dual-discrete (T2d)		
		n	Mean MI	k^* (%)	n	Mean MI	k^* (%)
Conspiracy	4	168	2.22	40.5	159	3.17	59.7
Climate action	3	117	0.17	0.0	126	1.05	0.8
Science and environment	5	195	0.11	0.0	193	0.87	0.0

of a rewritten text. Across all scored events at lineage rewrite depth one, mean conspiracy MI was 1.66, 2.97, and 2.18 under the three readings, compared with 0.13, 1.07, and 0.17 for climate action and 0.06, 0.79, and 0.07 for science and environment. Conspiracy events at depth two or greater averaged 0.69, 0.32, and 0.20 points more, respectively. These are event-weighted comparisons of differently selected groups, not paired increments along the same trajectories.

The early differences are consistent with the prompt manipulation. Conspiracy prompts instruct inversion or omission of claims that two articles’ audit items test, whereas the other homogeneous prompts instruct preservation (Section 4.3). The result therefore supports prompt responsiveness and the direction of the instrument, without establishing how human conspiracy communities reason or isolating network amplification.

5.3.2 Heterogeneous compositions

Named mixes differed far more than the pooled He label suggests (Table 5.4). `mix_00` carried four conspiracy seats in every condition. `mix_01` declared two but, like every mix except `mix_02`, realised four in the three cluster-assigned conditions (Table A.5). Pooling these two conspiracy-heavy mixes gave means of 1.62 and 2.95 under continuous and dual-discrete scoring. The conspiracy-free `mix_02` averaged 0.26 and 1.19, with no crossing under either instrument. This mix contrast exceeded the pooled arm difference under both instruments.

The aggregate mix contrast had article-specific exceptions (Figure 5.4). `mix_00` exceeded 3.5 on the consensus article under both instruments, whereas `mix_02` reached at most 0.90 on any article under continuous scoring. Under dual-discrete scoring, however, the polar-bear means reversed the aggregate ordering: 1.09 ($n = 8$), 1.40 ($n = 6$), and 1.57 ($n = 5$) for `mix_00`, `mix_01`, and `mix_02`. The corresponding SAI means were nearly flat at 1.15 ($n = 5$), 1.12 ($n = 7$), and 1.15 ($n = 7$). The evidence supports an aggregate composition contrast whose direction depends on the article.

The intermediate mixes did not order monotonically by declared conspiracy seats (Table 5.4). For example, `mix_03`, with one declared conspiracy seat, exceeded `mix_04`, with two, under both instruments. Mean MI and endpoint classifications also differed: `mix_00` exceeded `mix_01` in mean MI under both instruments but had a dual-discrete k^* rate 1.0 percentage point lower. Declared prevalence thus distinguished the aggregate extremes more clearly than the middle. The four mixes whose conspiracy-seat counts

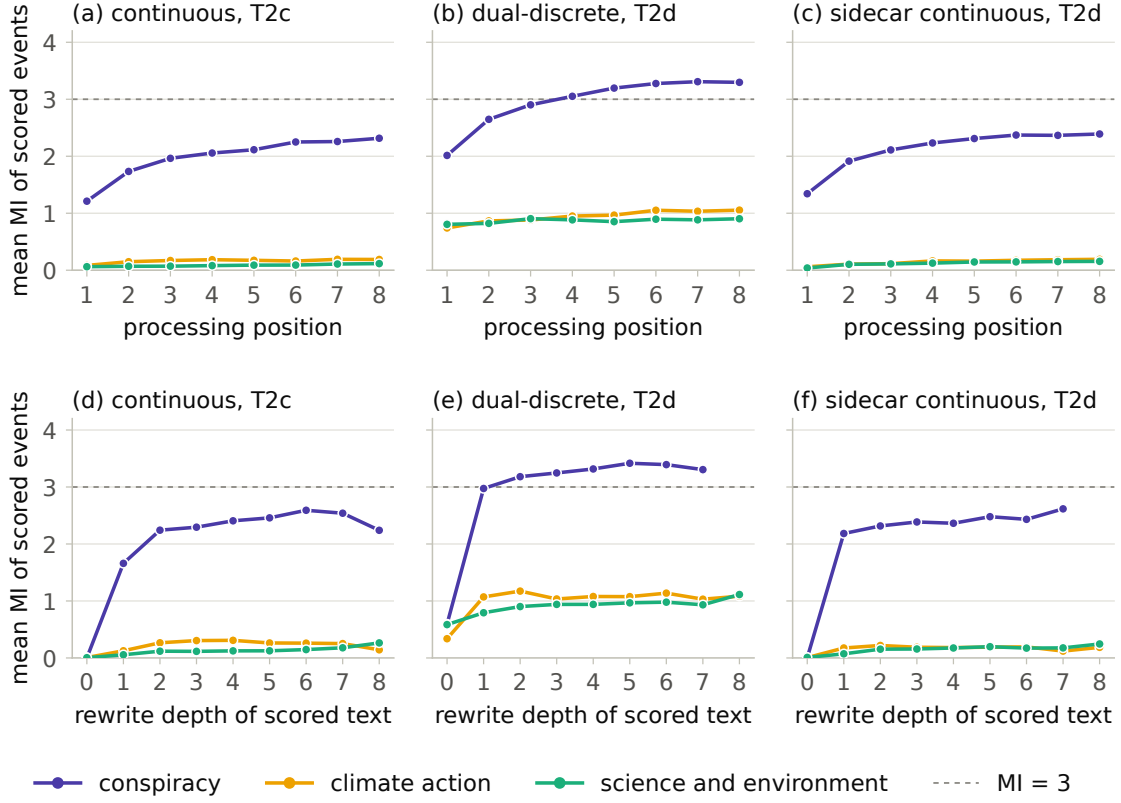


Figure 5.3: Mean MI of scored events in live homogeneous cells by family, by one-based processing position (top: recorded input hop count plus one, with position 1 denoting seed processing) and rewrite depth (bottom: depth 0 is unchanged source text), one column per instrument reading; rewrite depths 0–8 are shown, with points based on fewer than 20 events suppressed; dashed line at $MI = 3$. Conspiracy events at lineage rewrite depth 1, including forwarded copies, average 1.66, 2.97, and 2.18, against 0.00, 0.59, and 0.00 at depth 0 and 2.35, 3.29, and 2.39 at depths 2 or greater. These event means differ from the one-earliest-rewrite-per-cell percentages in the text.

Table 5.4: The six heterogeneous mixes collapsed across conditions and articles (48 cells per mix and instrument): declared composition, live cells n , live-cell mean MI, and k^* rate; every mix except `mix_02` realises four conspiracy seats on the three cluster-assigned conditions (Table A.5). The conspiracy-heavy mixes lead and `mix_02` trails.

Mix	Composition	Continuous (T2c)			Dual-discrete (T2d)		
		n	Mean MI	k^* (%)	n	Mean MI	k^* (%)
<code>mix_00</code>	4 conspiracy + 4 other	39	1.77	12.8	40	3.13	57.5
<code>mix_01</code>	2 conspiracy (declared) + 6 other	41	1.48	9.8	41	2.78	58.5
<code>mix_02</code>	0 conspiracy + 8 other	41	0.26	0.0	41	1.19	0.0
<code>mix_03</code>	1 conspiracy (declared) + 7 other	39	0.89	10.3	36	2.04	33.3
<code>mix_04</code>	2 conspiracy (declared) + 6 other	38	0.70	5.3	39	1.69	20.5
<code>mix_05</code>	2 conspiracy (declared) + 6 other	40	1.01	7.5	43	1.67	23.3



Figure 5.4: Live-cell mean MI by heterogeneous mix and article, both instruments (up to eight live cells per grid cell; none empty). Aggregate mix differences have article-specific exceptions: the dual-discrete polar-bear ordering reverses, and SAI shows little separation among the first three mixes.

increased each realised four such seats in three conditions, with different persona identities and placements (Section 4.3).

5.3.3 Article identity

Article identity ordered the cells similarly across arms and instruments (Table 5.5). Three slices shared the same full ordering; dual-discrete He differed in two adjacent pairs. The consensus article ranked highest in every slice. Polar bears ranked lowest except in dual-discrete He, where SAI scored 1.12 and polar bears 1.14. These pooled patterns allow article-by-composition interactions. Each article combines topic, wording, and five Boolean items, so the differences cannot isolate valence or truth status. The discrete polar-bear, SAI, and Paris means also include the unchanged-source errors identified in Section 5.2.1.

5.4 RQ3: network conditions

RQ3 asks how the eight bundled network conditions differ in same-regime H–He contrasts, live-cell mean MI, and scorable-cell survival within the eight-tick horizon. Each condition bundles a graph generator with an action policy and an assignment procedure, so the differences below describe conditions, not topologies (Section 4.4). The section reports the arm contrast C1, condition levels and survival with the exposure policy that shaped them, composition- and article-resolved views, k^* by condition, and robustness.

Table 5.5: Article-level live-cell mean MI and k^* rate by arm and instrument, collapsed across personas or mixes and conditions (37 to 88 live cells per entry). The consensus article leads in every slice.

Article	Homogeneous (H)				Heterogeneous (He)			
	Mean MI		k^* (%)		Mean MI		k^* (%)	
	T2c	T2d	T2c	T2d	T2c	T2d	T2c	T2d
scopex_2017	0.73	1.54	1.1	24.4	0.77	2.37	0.0	42.1
chemtrails_gates_2018_2021	1.35	2.14	29.3	32.9	1.79	2.39	10.0	38.1
sai_geoengineering	0.20	1.36	0.0	1.3	0.15	1.12	0.0	0.0
paris_agreement	1.34	1.73	24.7	25.7	1.68	2.46	16.2	51.3
climate_consensus	1.59	2.26	31.9	35.0	1.89	2.90	21.6	55.8
polar_bears	0.02	1.04	0.0	1.2	0.04	1.14	0.0	2.5

5.4.1 C1: the same-instrument arm contrast, condition levels, and survival

The same-instrument arm contrast C1 changed sign across conditions under continuous scoring but was positive in all eight under dual-discrete scoring (Table 5.6). Under continuous scoring, He exceeded H in the small-world, echo-chamber, polarised, and hierarchical conditions and fell below H in the other four. The positive differences outweighed the negative differences in aggregate, giving a condition-equal difference of +0.17. Weighting live cells equally gave a pooled difference of +0.15. Under dual-discrete scoring, the condition differences ranged from +0.16 (echo chamber) to +0.96 (hierarchical), with a pooled difference of +0.40 under both weightings.

Restricting the pooled contrast to runs with no HTTP error changed the continuous sign (last row of Table 5.6). Among runs without logged HTTP errors the continuous means were 0.95 (H) and 0.92 (He), a difference of -0.03 , whereas the dual-discrete means were 1.76 and 2.23, a difference of +0.47. The positive full-campaign continuous difference is therefore sensitive to this restriction; the sign reversal is a descriptive finding rather than evidence of a stable arm ordering. The dual-discrete difference kept its sign under this and the other restrictions in Section 4.11. These selected populations differ in more than HTTP-error status, so the sensitivity does not isolate an effect of scoring failure.

The H condition means vary little under both instruments, whereas the He series rises on the three cluster-assigned conditions under the continuous instrument (Table 5.6). The He series peaks at the hierarchical condition under both instruments, although under the dual-discrete instrument the echo-chamber and polarised He means fall below the chain and ring; the signed differences are positive in four of eight continuous comparisons and all eight dual-discrete comparisons. The higher continuous He means occur on the three cluster-assigned conditions, where persona placement also changes (Section 5.3). Their arm contrasts therefore combine composition, assignment, and graph structure.

Homogeneous condition means spanned 0.20 under each instrument. Heterogeneous means covered wider ranges: 0.66–1.52 under continuous scoring and 1.88–2.59 under dual-discrete scoring (Table 5.6). Hierarchical He ranked highest under both instruments; polarised He ranked second under continuous scoring but lowest under dual-discrete

Table 5.6: C1: live-cell mean MI of the H and He arms by network condition and instrument, with $\Delta = \text{He minus H}$, live over total cells, and pooled contrasts in the last rows. The zero-HTTP-error row gives live counts only. Differences use unrounded means. The continuous Δ changes sign; the dual-discrete Δ is positive throughout.

Condition	Continuous (T2c)					Dual-discrete (T2d)				
	Live H	Live He	H	He	Δ	Live H	Live He	H	He	Δ
Linear chain	62/72	32/36	0.91	0.86	−0.05	68/72	33/36	1.61	2.09	+0.47
Ring	65/72	33/36	0.92	0.88	−0.04	61/72	33/36	1.64	2.18	+0.54
Random ER	57/72	29/36	0.87	0.66	−0.21	56/72	29/36	1.78	2.04	+0.26
Small world	56/72	31/36	0.72	0.75	+0.03	59/72	32/36	1.60	1.91	+0.31
Scale-free	60/72	30/36	0.92	0.82	−0.10	57/72	29/36	1.70	1.99	+0.29
Echo chamber	63/72	32/36	0.85	1.38	+0.53	61/72	30/36	1.80	1.96	+0.16
Polarised	65/72	25/36	0.84	1.41	+0.58	57/72	26/36	1.70	1.88	+0.18
Hierarchical	52/72	26/36	0.88	1.52	+0.65	59/72	28/36	1.63	2.59	+0.96
Cell-pooled	480/576	238/288	0.86	1.02	+0.15	478/576	240/288	1.68	2.08	+0.40
Condition-equal			0.86	1.04	+0.17			1.68	2.08	+0.40
Zero-HTTP-error pooled	247	150	0.95	0.92	−0.03	304	139	1.76	2.23	+0.47

scoring.

These two He conditions also had the fewest live cells: 26 hierarchical and 25 polarised cells under continuous scoring, and 28 and 26 under dual-discrete scoring. Chain and ring lost the fewest cells and shared a rewrite-heavy action policy. Because the ring’s return is not processed within eight ticks, those two conditions expose the same processing path; their score differences do not demonstrate an effect of cyclic return (Section 4.4).

The inbox-admission audit confirms that node order shaped exposure in the dense conditions (Section 4.5.1). Across 49,817 recipient–tick slots, the admitted count always equalled the smaller of four and the proposed count. In all 27,777 capacity-bound slots (55.8 per cent), the admitted senders were the first four in node order. The rule discarded 172,453 of 324,580 proposed deliveries (53.1 per cent), with none discarded in the chain or ring. Rejection rates rose from 0.0 per cent for `node_0` to 88.8 per cent for `node_7`.

Sender family and node order were associated under the realised assignments. Conspiracy senders consequently lost a smaller share of proposed deliveries than science and environment senders in every dense He condition; in the echo chamber, the shares were 46.6 and 62.1 per cent (Table D.3). These exposure differences accompany the assigned compositions, so an arm or condition contrast cannot be attributed to graph edges or persona interaction alone.

5.4.2 Composition- and article-resolved views

The conspiracy family had the highest homogeneous mean in all eight conditions under both instruments. Its condition means ranged from 1.91 (small world) to 2.48 (polarised) under continuous scoring and from 3.00 to 3.49 under dual-discrete scoring. Science and environment means remained between 0.00 and 0.19 under continuous scoring. The companion archive supplies the condition-resolved heatmaps.

Mix orderings were less consistent. Either `mix_00` or `mix_01` ranked highest in seven conditions under each instrument; `mix_02` ranked lowest in five under continuous and

four under dual-discrete scoring. The exceptions to the highest-mix pattern occurred in cluster-assigned conditions: `mix_04` led polarised He under continuous scoring (2.22), and `mix_03` led hierarchical He under dual-discrete scoring (3.64). These assignments also depart from the declared seat orders (Section 4.3), preventing attribution to graph structure alone.

Some article extremes recurred across conditions, while the highest-scoring condition varied across articles (condition-resolved values are retained in the companion archive). Under the continuous instrument `polar_bears` was the lowest article in seven of eight conditions in both arms, and `climate_consensus` was highest in four conditions per arm with `paris_agreement` highest in three; under the dual-discrete instrument the highest article varied more. The hierarchical He column held the highest condition–article cells in the campaign, 3.75 (continuous) and 4.81 (dual-discrete) on `climate_consensus`. No common stability measure was computed for the two factors, so this is an observation about extremes, not a ranking of article identity above network condition.

5.4.3 Within-window k^* by condition

Both the instrument and the network condition moved the k^* rates substantially. H rates lay in a narrow band under both instruments, 11.5 to 16.7 per cent under continuous and 14.7 to 23.7 per cent under dual-discrete scoring, whereas He rates spanned 0.0 (linear chain and random ER) to 30.8 per cent (hierarchical) under continuous and 23.3 (echo chamber) to 42.9 per cent (hierarchical) under dual-discrete scoring. Every He condition had a higher rate under the dual-discrete instrument, by 12.1 (hierarchical) to 36.4 (ring) percentage points, yet the He between-condition range under continuous scoring (30.8 points) exceeded the pooled between-instrument shift (24.5 points). These descriptive comparisons do not establish a general ordering of factor importance.

5.4.4 Robustness to scoring coverage and window

The robustness analysis used three live-cell populations: all 1,436 live cells, the 840 live cells from 167 runs without HTTP errors, and the 1,125 live cells in which every audit-eligible event received a score (Table D.1). These restrictions also changed condition coverage. In particular, the zero-HTTP-error subset contained no dual-discrete He cell from the polarised, echo-chamber, or scale-free conditions.

The family ordering, `mix_00`–`mix_02` contrast, He reversal of the k^* rate between instruments, and dual-discrete arm difference retained their directions in all three populations. The continuous arm difference changed sign in the zero-HTTP-error subset. Article orderings also varied: SAI was below polar bears in full-population dual-discrete He (1.12 against 1.14), while chemtrails displaced consensus at the top of continuous He in both restricted populations (1.79 against 1.69 and 1.89 against 1.63).

Same-event condition-rank correlations (Section 5.2) were also sensitive to cell selection. The H and He correlations rose from 0.62 and 0.12 to 0.90 and 0.70 in the zero-HTTP-error subset, and to 0.95 and 0.83 in the fully scored subset. These changes cannot separate the contributions of missing scores, condition composition, and scoring specification.

The eight-tick window truncated most cascades (Figure D.1). Of the 1,436 live cells, 1,334 (92.9 per cent) still produced events at tick 8, including every live cell of the small-world, scale-free, echo-chamber, and polarised conditions, against 80.5 per cent of chain and 79.2 per cent of ring cells. The median live cell contained 44 text-changing rewrites in total. The median per-cell maximum lineage depth was six rewrites. Live-cell means rose monotonically as the window extended from two to eight ticks (dual-discrete H: 1.37, 1.57, 1.64, and 1.68) and the dual-discrete conspiracy family crossed 3 only once the window included ticks 5 and 6, so convergence of the pooled means was not established, whereas the family ordering was the same at the four tested truncation points under every reading (Chapter D).

5.5 RQ4: the D-net stress test

RQ4 asks what the 63-node constructed hashtag-graph stress test can establish about recurring composition patterns and where that comparison fails. D-net has 63 nodes, 228 directed edges (171 unique undirected pairs), and one fixed seed node (`user_chemtrails_hub`); it was run in four configurations crossing the two instruments with a conspiracy-homogeneous assignment (all 63 seats `conspiracy_believer`) and a mixed assignment of 28 conspiracy, 19 climate-action, 13 environmental, and 3 expert seats, 56 of 63 with the exact intended persona identifier (Section 4.13). Each configuration propagated two articles, which gave eight cells, all live.

The homogeneous conspiracy assignment scored above the mixed assignment in all four article–instrument pairs (Table 5.7). Article ordering depended on the instrument: chemtrails scored higher under continuous scoring, whereas SCoPEX scored higher under dual-discrete scoring in both assignments. The dual-discrete headline exceeded the continuous-campaign headline in three of four comparisons; the mixed chemtrails cell was the exception. The sidecar lay below the headline in all four dual cells.

The D-net endpoints also differed between instruments. A within-window k^* occurred in three of four dual-discrete cells and one of four continuous cells, consistent with the direction reported in Section 5.2. In every retained cell, 61 of 63 nodes received a message within eight ticks. The eight cells from the other execution of each configuration entered only the archived structural check (Chapter B).

The D-net ordering is consistent with the named composition comparison in Section 5.3. Its homogeneous assignment contains only conspiracy personas, whereas the mixed assignment contains 28 conspiracy seats among 63. The grid’s H arm instead pools three families, so its pooled H–He ordering addresses a different comparison. The larger-graph result is a descriptive recurrence on one graph, not an independent replication of the grid: graph size, scored-event counts (407 to 1,076 per cell), and article coverage differ.

The archived structural comparison contained an extraction error. It used a sender’s sender as the parent, omitted first-hop edges, and sometimes selected the wrong root. It also pooled two executions of each configuration. Chapter B preserves the superseded results and documents the corrected reconstruction on the eight retained cells.

The corrected first-receipt trees have mean depth 4.25, breadth 33, and structural

Table 5.7: Article-level auditor scores in the four D-net configurations: headline live-cell mean MI, continuous sidecar on the same events in dual runs, scored events, and first within-window k^* tick. The conspiracy-homogeneous assignment exceeds the mixed one in every pair.

Assignment	Instrument	Article	Headline	Sidecar	n scored	k^*
H conspiracy	continuous	SCoPEX	1.73	—	970	none
H conspiracy	continuous	chemtrails	3.34	—	984	tick 2
He mixed	continuous	SCoPEX	1.00	—	591	none
He mixed	continuous	chemtrails	2.51	—	1,035	none
H conspiracy	dual-discrete	SCoPEX	4.25	2.25	1,076	tick 2
H conspiracy	dual-discrete	chemtrails	3.73	3.13	749	tick 4
He mixed	dual-discrete	SCoPEX	2.69	1.13	407	tick 7
He mixed	dual-discrete	chemtrails	1.53	1.15	1,027	none

virality 3.17, within the chosen 30 per cent tolerance of the reference values 4, 33, and 2.70. The reconstruction uses the sender of each node’s first event as its parent and fixes the seed as the root. Six trees reach depth four and two reach depth five; Section B.1 also reports the input graph and assignment diagnostics.

The reference nevertheless comes from the simulation’s own input edges. Its depth is the seed’s eccentricity, its breadth the seed’s out-degree, and its virality value the mean contact-graph distance. The close match is therefore consistent with first receipts following short paths on the supplied graph. It provides no independent evidence of Twitter fidelity. The stress test’s substantive result remains the recurrence of one composition ordering on a larger constructed graph.

5.6 Answers supported by the campaign

RQ1 identifies scoring sensitivity and a discrete unchanged-source defect. RQ2 identifies expected prompt-family contrasts, with article-specific exceptions among mixes. RQ3 identifies differences between bundled protocols whose interpretation depends on scorable-cell selection and admission order. RQ4 finds a recurring composition ordering on one constructed graph, while its structural comparison supplies no independent platform validation. These results support a conditional account of the executed simulation; Chapter 6 examines competing explanations and the evidence needed for stronger claims.

Chapter 6

Discussion, Validity, and Future Work

The campaign’s strongest finding concerns measurement: the scoring rule and the selected analysis population change what appears to be a substantive propagation result. The composition patterns document prompted model behaviour, while the network and D-net comparisons remain conditional on bundled protocols. This chapter interprets those findings and identifies the evidence needed to go beyond them.

6.1 Measurement before substantive interpretation

The dual sidecar demonstrates that identical generated texts can receive materially different scores. The cell-equal signed gaps were 0.84 in H and 1.05 in He. Of these, 0.32 and 0.61 follow algebraically from the larger discrete omission penalty. The residual reflects differing item judgements across separate stochastic calls; it cannot be assigned entirely to the rubric or treated as a measure of true degradation. The same-model design makes this a comparison of two implementations of one model, not independent confirmation by two observers.

The unchanged-source control shows that nonzero scores need not reflect any textual change. The discrete auditor assigned nonzero scores to 38.4 per cent of unchanged-source audits in T2d, compared with 1.0 per cent for their continuous sidecars. Such events cannot establish loss during propagation because the text did not change. Errors concentrated on particular negatively keyed items, especially the polar-bear article, so subtracting a common baseline from all rewritten texts would be unjustified. The zero-recoding sensitivity changes some condition rankings but does not calibrate the instrument.

The continuous instrument returned zero on a larger proportion of unchanged-source audits. This result does not establish its accuracy on omissions, contradictions, or novel additions. Independent human annotations of altered texts remain necessary. In measurement-theoretic terms, the audit narrows the admissible inference rather than establishing criterion validity (Cronbach and Meehl, 1955; Messick, 1995). Automated judge agreement alone cannot supply that missing criterion (Zheng et al., 2023).

The threshold results also depend on the instrument. Omitting all five items yields a discrete score of five but a continuous score of 2.5. Applying the same cut at three therefore tests different mixtures of omission and contradiction. Moreover, k^* exists exactly when the mean MI at the last scored tick exceeds three. Its occurrence indicates an endpoint within the observed window; it neither establishes persistence beyond eight

ticks nor identifies a firestorm. The reported rates remain useful as a sensitivity diagnostic if those definitions accompany them.

6.2 Composition and the meaning of the arm contrast

The family ordering largely follows instructions in the prompts. Conspiracy prompts explicitly direct omission or inversion of some source claims, whereas the other homogeneous prompts direct preservation. High scores already occur on the first text-changing rewrite. Later rewrites show higher average scores, but those comparisons use differently selected event groups. The campaign therefore supports prompt responsiveness, without isolating incremental network amplification.

The pooled H–He comparison obscures this family-level pattern. H combines four conspiracy, three climate-action, and five science and environment profiles; it is not synonymous with an all-conspiracy society. He combines six named mixes whose realised shares and placement vary by condition. Its pooled live-cell mean exceeds the pooled H mean under both instruments, yet the homogeneous conspiracy family exceeds He under both instruments. An unqualified claim that diversity raises or lowers degradation would conflate these compositions.

The mixes make the distinction visible. The two conspiracy-heavy mixes have higher aggregate means than the conspiracy-free mix, but the contrast is not uniform across articles. Under dual-discrete scoring, the polar-bear means reverse the aggregate ordering, and the SAI means are nearly flat (Figure 5.4). They also occur in a regime whose unchanged-source errors vary by article.

D-net changes the meaning of H again: all 63 seats hold `conspiracy_believer`. Its mixed assignment contains 28 conspiracy seats among 63, compared with a realised conspiracy share of 29.9 per cent in the grid’s He arm. The higher homogeneous D-net scores therefore do not contradict the pooled grid result. They compare a different pair of compositions. Isolating diversity would require fixed family shares, matched seed and hub positions, and a controlled change in the number of distinct profiles within each family.

6.3 Network conditions and scorable-cell survival

The network labels identify bundled protocols rather than isolated graph treatments. Chain and ring use a distinct action policy, with a trust threshold lower than in the other conditions. Three conditions also change persona assignment. The inbox cap further changes effective exposure: the first four messages in sender order are admitted, and 53.1 per cent of proposed deliveries are discarded. In heterogeneous dense graphs, this rule favours earlier seats and can favour particular families through placement. A difference between condition means cannot be attributed to edges alone.

The eight-tick horizon restricts the mechanisms observed. A message traversing the ring would return for processing at the seed only at tick nine; all retained ring runs therefore lack this cyclic return. Chain and ring expose the same available processing

path in the observed window. Their score differences concern different realised actions, generations, and audits, rather than a demonstrated effect of feedback around a cycle.

Scoring availability introduces another source of selection. Under continuous scoring, the pooled He minus H difference changes from +0.15 to -0.03 in the zero-HTTP-error subset. Under dual-discrete scoring it remains positive, changing from +0.40 to +0.47. The selected continuous He subset has 150 live cells across seven conditions, with echo chamber absent; the dual-discrete He subset has 139 across five, with echo chamber, polarised, and scale-free absent. These restrictions change condition and time composition, so they diagnose sensitivity rather than correct an API effect.

Live-cell means also condition on at least two scored events. A cell may fail that rule because propagation stops or because its audits fail. Scorable-cell survival must therefore accompany severity, and should not be called cascade survival without qualification. The hierarchical He condition combines high conditional scores with 26 live cells under continuous and 28 under dual-discrete scoring. No analysis of retained scores alone can recover the unobserved trajectories or verdicts of excluded cells.

The shortened-window analysis strengthens the horizon qualification. Most live cells still produce events at tick eight, and aggregate means change as the window extends. Stability of the family ordering over the tested windows is useful descriptive evidence, but it does not establish convergence or equivalence to the prior 30-hop design. Eight ticks were a resource constraint, not an empirically validated compression of 30 hops.

6.4 D-net and the boundary of external validation

The larger graph supplies a bounded recurrence: the homogeneous conspiracy assignment exceeds the specified mixed assignment in four of four article-instrument comparisons. Those observations concern four configurations on one constructed graph, with two articles each. They are not independent replications across real networks and do not isolate diversity from composition, trust, or realised exposure.

The structural audit changes the interpretation of a previously reported fidelity check. The archived extractor used a sender's sender as the parent, omitted first-hop edges, and sometimes selected the wrong root. Correcting those errors yields mean depth 4.25, breadth 33, and structural virality 3.17 for the eight retained first-receipt trees, within the chosen 30 per cent tolerance band around the respective reference values of 4, 33, and 2.70. The reference, however, is calculated from the simulation's own input graph. First receipts often follow short paths on that graph, although two corrected trees reach depth five. The close match is consistent with propagation along the supplied graph, but these aggregate statistics alone do not verify individual deliveries or establish similarity to independently observed Twitter cascades.

The data provenance reinforces that boundary. D-net nodes are hashtag-defined synthetic roles, and most edges arise from documented construction rules. Counts from the first 5,000 OSF records annotate 48 existing arcs without changing the edge set. The result is empirically informed, but it is neither an observed account network nor a timestamped propagation corpus. A failure or pass against this reference cannot resolve

external validity (Sargent, 2013; Larooij and Törnberg, 2025).

6.5 Relation to the literature and Pfeffer’s framework

The prior auditor–node study supplies the method rather than a directly comparable benchmark (Maurya et al., 2025). It used ten items, general-news articles, and 30-hop chains; the present study changes all three and adds an alternative auditor. Its contribution is the audit of that transfer, including a negative control that exposes a scoring defect. It does not establish that differences from the prior study arise from climate subject matter or network complexity alone.

Transmission-chain studies show that LLM rewriting can selectively retain content (Acerbi and Stubbersfield, 2023), compress a story mainly at its first retelling (Breithaupt et al., 2024), and approach model- and instruction-dependent attractors (Perez et al., 2025). These findings make ordinary rewriting a relevant alternative explanation for apparent propagation effects. Here, family differences already appear near the first rewrite under prompts that instruct preservation or distortion. Without a neutral-rewrite baseline, the campaign cannot isolate an additional network effect beyond such prompted transformations. Its eight-tick window also cannot establish long-run convergence.

The audit adds an evaluator-level distinction to this literature. FActScore measures the proportion of generated atomic facts supported by a reliable source (Min et al., 2023); MI instead scores recoverability of five selected source claims. It can therefore penalise faithful compression and miss false additions outside those items. Same-text comparisons and unchanged-source controls show that the evaluator can change assessed loss without any change in the text. This finding makes instrument checking a prerequisite for interpreting the present campaign’s propagation scores.

Pfeffer’s framework was applied as a mapping from theory to implemented design (Table 3.1), including the constructed D-net diagnostics. The study varies bundled proxies for clusters and diversity and represents some information-flow constraints. It does not independently manipulate all seven factors, measure negative affect, or analyse a sudden volume surge. It consequently neither confirms nor refutes firestorm theory. Testing the framework against observed propagation on real complex networks remains future work requiring independent empirical cascades and appropriate escalation outcomes.

6.6 Validity and reproducibility

Three limitations require new evidence. First, one retained run per configuration provides no repeat-run distribution. The 1,728 cells share configurations and articles; treating them as independent replications would overstate precision. Sign counts and rank correlations describe this campaign only. Archived reruns were operational replacements, not a prespecified replication sample.

Second, the instrument lacks independent criterion evidence. Both judges share the generator’s model family, item selection is author-curated, and some persona prompts explicitly name the claims that agents are instructed to distort. Five fixed questions

cannot cover every possible false addition. The unchanged-source control detects one error mode but cannot validate altered-text scores. Article differences also combine topic, wording, and item difficulty.

Third, the simulation has narrow external scope: one model alias, six articles, synthetic personas, eight-node graph realisations, and one larger constructed graph. No human belief, persuasion, affect, or platform response is measured. Results apply to the configured system and do not warrant general claims about real communities.

Several additional threats are partially measurable from the archive. API failures affect both generated text and scoring coverage: rewrite failures propagate unchanged text, while audit failures remove observations. Either distortion or correction could be suppressed by an unchanged-text fallback, so its bias has no fixed direction. The auditor's response parser defaults to fully correct verdicts when it cannot parse a judge reply. No such warning was observed, but raw judge responses were not retained. Node-order admission, realised persona shares, and window truncation can be reconstructed, yet their causal effects require controlled reruns.

The analysis fixes run selection through a manifest and uses event-reconciled tables. Two cells retain only mid-audit histories; their completed aggregate summaries exist, but 220 completed event-level scores do not. The thesis reports the results that can be regenerated from the retained records (Section C.3). The companion archive includes the reconciliation additions, verification scripts, and checksums; the earlier repository commit alone does not contain them.

This supports reproducibility of the archived analysis, not exact replay of the simulation. Action and some graph randomness were unseeded; no API seed, response fingerprints, or raw judge replies were archived. A fresh execution would be a protocol replication (Richiardi et al., 2006). The large number of model calls also creates an access and cost constraint, analogous to the resource inequalities discussed in platform-data research (Ruths and Pfeffer, 2014).

6.7 Ethics and responsible interpretation

The simulation recruited no human participants and used authored stimuli, synthetic prompts, and generated rewrites. The separate OSF preprocessing did transiently process public tweet text and user fields to extract hashtag aggregates. Those fields were discarded; the retained outputs contain aggregates and an identifier-format diagnostic, including sampled identifier tokens. The platform records did not enter model prompts. No formal ethics determination is documented in the supplied study materials, and no institutional exemption is asserted.

Persona names should not be read as validated descriptions of real participants. The explicit distortion instructions make the model behaviour partly a design choice and can reinforce caricatures if this boundary is omitted. Complete prompts are retained for reproducibility, while the thesis attributes findings to configured models. The campaign does not validate a strategy for influencing real users, although simulations of rewriting have potential adversarial uses (Lazer et al., 2018).

A successor involving human coders should obtain the applicable institutional determination before recruitment and document consent, blinding, and exposure safeguards. Such safeguards matter because misinformation can continue to influence reasoning after correction (Lewandowsky et al., 2012). Further platform-data collection would require a separate account of processing, access, and release conditions.

6.8 Priority future work

The next study should prioritise identification and measurement over a larger grid. Measurement validation comes first: use affirmative and negative controls, faithful compression, omissions, contradictions, and added false claims; obtain blinded, stratified human annotations; and compare judges from another model family. Version any revised rubric and reject malformed replies as missing. Neutral-rewrite and forward-only conditions, together with prompts that do not name audit items, would distinguish summarisation loss from instructed distortion.

Next, replicate a smaller set of configurations with recorded graph and action seeds, multiple generation runs, and uncertainty summaries at the run level. Hold action and trust rules constant across graphs, counterbalance seed and hub identities, vary placement independently of family shares, and randomise or counterbalance admission order while logging rejected deliveries. Prespecify the estimand and report propagation, scoring availability, and conditional severity separately.

Extend the horizon beyond cyclic return paths and compare it with a matched 30-hop chain baseline. Shortened-window stability should be assessed alongside trajectory and stopping diagnostics, rather than assumed to imply convergence. Logging raw judge replies, model metadata, and final event histories would also prevent the present ambiguity between completed summaries and incomplete archived scores.

Finally, validating simulated propagation against platform dynamics requires several independent, timestamped empirical cascades and a reference not used to construct the simulation input. A test of Pfeffer’s framework would additionally need defined volume and affect outcomes and separate manipulations of its factors. These steps would turn the current design audit into a stronger test of propagation mechanisms.

Chapter 7

Conclusion

This thesis adapts an inherited auditor–node framework to climate and geoengineering messages and audits its outputs. The campaign comprises 1,728 configuration–article cells across eight network conditions, with a separate stress test on a constructed 63-node graph.

RQ1 identifies instrument sensitivity. On identical generated texts, discrete scores exceeded continuous scores by 0.84 in H and 1.05 in He, weighting cells equally. Larger omission penalties explain part of these gaps. Unchanged-source controls exposed nonzero scores without textual change. Threshold classifications and some condition orderings depend on the scoring procedure; neither procedure is independently validated on altered texts.

RQ2 identifies differences between prompt configurations. The conspiracy family had the highest aggregate scores and within-window k^* rates of 40.5 per cent under continuous and 59.7 per cent under dual-discrete scoring. Only one live cell from the other homogeneous families had a defined k^* . Conspiracy-heavy mixes exceeded the conspiracy-free mix in aggregate mean MI, with article-specific exceptions; the conspiracy-free mix had no k^* . These results support prompt responsiveness without isolating a causal diversity effect.

RQ3 identifies differences between bundled protocols. He mean MI exceeded H in four of eight continuous and all eight dual-discrete comparisons. Hierarchical He had the highest mean under both instruments; polarised He had the lowest scorable-cell survival. Restriction to runs without logged HTTP errors reversed the pooled continuous arm difference. Joint variation in graph, policy, placement, and scoring availability prevents an isolated topology ranking.

RQ4 identifies recurrence on one larger constructed graph: the homogeneous conspiracy assignment exceeded the mixed assignment in all four article–instrument comparisons. The corrected structural statistics use the simulation’s input graph as their reference. They therefore supply no independent evidence of Twitter fidelity.

The contribution is the demonstrated sensitivity of this campaign’s conclusions to its instrument and analysis population. The architecture and indices remain prior co-authored work (Maurya et al., 2025). Independent annotations and controlled replications are needed to validate altered-text scores and isolate propagation mechanisms. The present audit establishes why instrument checks, explicit scoring populations, and realised exposure must accompany interpretation of its degradation scores.

Appendix A

Measurement and Executed Design

This appendix retains the item keys, scoring prompts, and design details needed to assess the central results. Full persona prompts, JSON schemas, pilot results, secondary heatmaps, and historical diagnostics are supplied in the companion archive rather than repeated in the printed thesis.

A.1 Audit items

The 30 audit items are the questions on which every MI value in this thesis rests, so they are reproduced verbatim with their expected answers and the span of the source text on which each answer rests. Table A.1 lists them by article; the supporting spans are quoted from the article text in `articles/merged.json` and were checked programmatically to occur verbatim there. Seventeen items expect Yes and thirteen expect No (three Yes and two No for every article except `sai_geoengineering`, which expects two Yes and three No); the articles are 169 to 229 words long. Every No item negates a misinformation claim (for example, whether a stable subpopulation count proves that sea-ice loss is not a threat), so an omission of the target claim and an accurate denial of it are difficult for a judge to separate from the text alone. Every failure of the unchanged-source control (Section 4.6.1) fell on a No item: under the discrete instrument, verdicts of 0 or -1 on byte-identical source texts occurred on `polar_bears` items 3 and 5, `paris_agreement` item 2, `sai_geoengineering` items 4 and 5, and `climate_consensus` items 3 and 4, and on no expected-Yes item under either instrument. The auditor scores the five items of an article as one summed index (Section 4.6).

Appendix A Measurement and Executed Design

Table A.1: The 30 audit items of the six core articles: question, expected answer, and the supporting span quoted from the source text. Generated by `notes/verify_supplementary.py` from `articles/merged.json`; bracketed ellipses mark omitted words between quoted source fragments.

Article	Item	Question	Answer	Supporting source span
scopex_2017	1	Was SCoPEX designed as a small scientific measurement experiment rather than a deployment of solar geoengineering to cool the planet?	Yes	not a test of deploying geoengineering and was not intended to cool
	2	Did the proposed science flight plan to inject thousands of tonnes of material into the stratosphere?	No	on the order of 100 grams to less than 2 kilograms
	3	Was the proposed release typically calcium carbonate in amounts under 2 kilograms?	Yes	typically calcium carbonate [...] less than 2 kilograms
	4	Did the 2021 Sweden platform test plan include releasing aerosols?	No	hardware test only [...] no particles were to be released
	5	By 2024, had Harvard’s principal investigator stopped pursuing SCoPEX?	Yes	In March 2024 Harvard announced that Keutsch was no longer pursuing SCoPEX
chemtrails_gates_2018_2021	1	Does the scientific community treat chemtrails as a verified secret spraying program?	No	The scientific consensus is that this is false
	2	Are persistent aircraft trails explained as ordinary ice-crystal contrails under certain atmospheric conditions?	Yes	ordinary contrails made of ice crystals that linger under certain humidity
	3	Did Debnath and colleagues analyze hundreds of thousands of #geoengineering tweets and find chemtrails conspiracy language dominant?	Yes	analyzed 814,924 English-language tweets [...] chemtrails language dominates
	4	Did public #geoengineering Twitter activity surge around the SCoPEX project around April 2017?	Yes	a peak near the April 2017 launch, on the order of 8,000
	5	Does Bill Gates funding climate or geoengineering research constitute evidence that aircraft are spraying chemicals for population control?	No	Funding [...] research is not evidence of a covert aircraft spraying program
sai_geoengineering	1	Is stratospheric aerosol injection a proposed research and assessment topic rather than an operational global cooling program already running via commercial jets?	Yes	a proposed solar radiation modification method [...] no operational global SAI deployment
	2	Would SAI remove carbon dioxide from the atmosphere and stop ocean acidification?	No	would not remove carbon dioxide [...] would not stop ocean acidification

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Appendix A Measurement and Executed Design

Table A.1 (continued)

Article	Item	Question	Answer	Supporting source span
	3	Did large volcanic eruptions such as Pinatubo 1991 produce temporary cooling via stratospheric aerosols, which SAI would try to mimic?	Yes	cooling after large volcanic eruptions such as Mount Pinatubo in 1991
	4	Does the IPCC treat SAI as having no side effects if deployed?	No	Risks discussed in assessments include ozone chemistry, changes in precipitation
	5	Is a small scientific plume experiment the same thing as planetary-scale SAI deployment?	No	measure aerosol microphysics in a tiny plume, not to cool the planet
paris_ agreement	1	Was the Paris Agreement adopted in 2015 with a goal of holding warming well below 2°C and pursuing 1.5°C?	Yes	adopted [...] on 12 December 2015 [...] well below 2 C [...] 1.5 C
	2	Does the Agreement assign each country a single fixed tonnage quota with no NDC updates?	No	a ratchet mechanism of successive NDCs rather than a single top-down allocation
	3	Are nationally determined contributions the main vehicle for Parties' mitigation pledges?	Yes	Parties submit nationally determined contributions (NDCs) describing their mitigation plans
	4	Does the Paris Agreement secretly authorise a global chemtrail spraying programme?	No	it is not a secret authorisation to spray the atmosphere
	5	Would meeting the temperature goals require deep cuts in greenhouse gas emissions rather than aviation spraying?	Yes	requires deep, sustained cuts in fossil CO2 [...] not covert aviation spraying
climate_ consensus	1	Do IPCC assessments and literature surveys find that humans are the dominant cause of recent global warming?	Yes	primarily caused by human greenhouse-gas emissions is documented in IPCC assessment reports
	2	Is the greenhouse effect of carbon dioxide established physics?	Yes	The greenhouse effect of CO2 is established physics
	3	Does a petition or 'European Climate Declaration' by mixed professionals overturn the assessed climate literature?	No	does not negate the assessed literature
	4	Is remaining uncertainty about climate sensitivity the same as uncertainty about whether CO2 causes warming?	No	Uncertainty remains about the exact magnitude of climate sensitivity [...] not about whether
	5	Have surveys of climate papers taking a position often found around 97% endorsement of anthropogenic warming?	Yes	about 97% endorsed anthropogenic global warming

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Table A.1 (continued)

Article	Item	Question	Answer	Supporting source span
polar_bears	1	Do polar bears rely on sea ice to hunt seals?	Yes	depend on sea ice as a platform for hunting seals
	2	Has Arctic sea ice declined over the satellite record as the Arctic warmed?	Yes	sea ice has declined in extent and thickness over the satellite era
	3	Does a stable or increasing count in some subpopulations prove that sea-ice loss is not a threat?	No	That snapshot does not cancel the physical dependence on ice
	4	Is habitat loss from declining sea ice considered a primary long-term threat by specialist assessments?	Yes	loss of sea-ice hunting habitat as the primary long-term threat
	5	Would an increase in one subpopulation show that anthropogenic warming is a hoax?	No	no scientific basis for the claim [...] global warming is a hoax

External references support the intended item keys, but do not constitute independent validation of the complete stimuli. For example, the IUCN Polar Bear Specialist Group’s summary of its 2015 assessment identifies sea-ice loss as the principal circumpolar threat, supporting polar-bear item 4 (IUCN Polar Bear Specialist Group, n.d.). The companion source ledger links each article’s items to the cited evidence and records unresolved attribution details. In particular, the archived chemtrails text names the Gates Foundation as a research funder, whereas the cited SCoPEX account does not establish that attribution (Keutsch Group, 2024). This unresolved detail is outside its five audit items; the executed stimulus and all reported scores are preserved.

A.2 Prompt templates and IFD fields

The prompt templates are the exact strings that produced every rewrite and every auditor score, so they are reproduced verbatim from the engine source. Each reinterpret action combines the selected persona’s `systemPrompt` with the fixed user template in Listing A.1; forward actions make no model call, and both agent and auditor outputs are model outputs rather than human ratings.

```
Here is the information you received. Rewrite it through your
perspective in 100-200 words. Return only the rewritten text,
no preamble:

<incoming content>
```

Listing A.1: Rewrite user prompt from `src/SimulationNode.js`. The placeholder in angle brackets is replaced by the incoming message text.

The two auditor prompt templates differ only in their system prompt and the closing query line of the user prompt. Listing A.2 reproduces both system prompts and the shared user template from `src/Auditor.js`; in the dual regime the engine issues the two calls in parallel on the same event.

```

DISCRETE SYSTEM PROMPT
You are a fact-checker. For each question, the expected correct
answer is provided.
Evaluate the given text and return a score for each question:
  1 = text correctly states the expected answer
  0 = text does not contain enough information to address the question
 -1 = text contradicts or distorts the expected answer
Return only a JSON object: {"answers": [1, 0, -1, ...]}

CONTINUOUS SYSTEM PROMPT
You are a fact-checker. For each question, the expected correct
answer is provided.
Rate how accurately the given text addresses each question on a
scale from 0.0 to 1.0:
  1.0 = text fully and correctly states the expected answer
  0.5 = text does not address this question (information is
        missing or neutral)
  0.0 = text directly contradicts or inverts the expected answer
Intermediate values represent partial accuracy (e.g. 0.8 = mostly
correct, 0.2 = mostly wrong).
Return only a JSON object: {"scores": [1.0, 0.5, 0.8, ...]}

USER PROMPT, BOTH INSTRUMENTS
Text:
<text>

Questions (with expected answers):
1. <question> (expected: Yes|No)
...

discrete closing line:
IFD_SCORE_QUERY: Return only the JSON with values 1, 0, or -1.
continuous closing line:
CONTINUOUS_SCORE_QUERY: Return only the JSON with float scores 0.0 to 1.0.

```

Listing A.2: Discrete and continuous auditor templates from `src/Auditor.js`. Angle brackets mark substituted text.

The Information Fidelity and Distortion (IFD) record stores the item vectors, MI, and ancillary rates. The reported analyses use MI, the continuous sidecar, their gap, and item-score correlation. Correctness, missingness, contradiction ratios, and entropy-like fields are stored but are not separately validated constructs or reported outcomes. The score definitions are in Section 4.6; the complete schema remains in the companion source.

The parse default is fail-open and must be recorded because it biases towards low scores. When the auditor response contains no parsable JSON object, the engine logs a warning and returns a score of 1 for every discrete item and 1.0 for every continuous item, that is, a fully correct verdict; the engine maps discrete values other than 1 and -1 to 0 and clamps continuous values to $[0, 1]$. No parse-fallback warning or payload anomaly was found in the retained records; because raw auditor replies were not archived, an unrecorded or structurally valid malformed reply cannot be excluded (Sections 4.11

Table A.2: Item-score confusion matrix on the same scored dual events: rows give the discrete verdict, columns the continuous score q binned as correct ($q \geq 0.67$), missing ($0.33 < q < 0.67$), or contradicted ($q \leq 0.33$). Live T2d cells only, all five items of every scored dual event. Generated by `notes/verify_supplementary.py` from the retained node histories.

	Continuous item bin			
Discrete verdict	correct	missing	contradicted	Total
<i>H arm: 34,388 scored dual events, 171,940 item pairs, exact agreement 81.7%</i>				
Correct (+1)	110,452	784	1,644	112,880
Missing (0)	10,287	9,299	3,015	22,601
Contradicted (−1)	14,908	840	20,711	36,459
Total	135,647	10,923	25,370	171,940
<i>He arm: 13,470 scored dual events, 67,350 item pairs, exact agreement 79.4%</i>				
Correct (+1)	39,368	503	498	40,369
Missing (0)	6,047	8,585	2,113	16,745
Contradicted (−1)	4,397	336	5,503	10,236
Total	49,812	9,424	8,114	67,350

and 6.6). A successor study should replace the default with a null score.

The two judges classified the same item identically in about four of five cases. Table A.2 cross-tabulates, for every item of every scored dual event in live T2d cells, the discrete verdict against the continuous score binned at the auditor’s own cut-points ($q \geq 0.67$ correct, $0.33 < q < 0.67$ missing, $q \leq 0.33$ contradicted). Exact agreement was 81.7 per cent of 171,940 item pairs in H and 79.4 per cent of 67,350 in He. The disagreement is asymmetric: the discrete judge was the harsher (missing or contradicted where the continuous judge said correct, or contradicted where it said missing) in 15.1 (H) and 16.0 (He) per cent of pairs, the continuous judge in 3.2 and 4.6 per cent; the single largest off-diagonal cell in H is a discrete −1 against a continuous score of at least 0.67 (14,908 pairs, 8.7 per cent). The continuous judge returned exactly 0, 0.5, or 1 for 99.8 per cent of item scores (0.8 or 0.9 in the remaining 0.2 per cent), so its item judgements were almost entirely three-valued in this archive, and the same-event gap consists of classification disagreement plus the weight of 0.5 rather than 1 on missing items.

A.3 Unchanged-source control by article

Table A.3 supplies the article-level denominators behind the control in Section 5.2.1. These are repeated audits within runs, not independent replications.

A.4 Topology generator definitions

The eight generators define the graph component of each bundled network condition, and their parameters are the ones the configuration files pass to `src/SocietyGraph.js`. Table A.4 lists each generator with its recorded parameters and its construction rule; the realised graph stored in `graph_topology.json` for each run, not the generator label,

Table A.3: Unchanged-source control by article and instrument across all retained cells, including dead cells: audits byte-identical to the curated article (n), number and share with $MI>0$, and mean MI ; the discrete headline and continuous sidecar score the same T2d events. The expected score is zero; nonzero scores are much more frequent under the discrete headline.

Article	Continuous headline (T2c)			Discrete headline (T2d)			Continuous sidecar (T2d)		
	n	$MI>0$	mean MI	n	$MI>0$	mean MI	n	$MI>0$	mean MI
SCoPEx 2017	992	0 (0.0%)	0.000	882	0 (0.0%)	0.000	882	0 (0.0%)	0.000
Chemtrails–Gates	1,013	0 (0.0%)	0.000	775	0 (0.0%)	0.000	775	0 (0.0%)	0.000
SAI geoengineering	839	6 (0.7%)	0.006	898	554 (61.7%)	0.912	898	6 (0.7%)	0.007
Paris Agreement	800	32 (4.0%)	0.031	747	655 (87.7%)	0.877	747	37 (5.0%)	0.036
Climate consensus	783	0 (0.0%)	0.000	529	11 (2.1%)	0.042	529	0 (0.0%)	0.000
Polar bears	918	0 (0.0%)	0.000	406	405 (99.8%)	1.628	406	0 (0.0%)	0.000
All six articles	5,345	38 (0.7%)	0.006	4,237	1,625 (38.4%)	0.509	4,237	43 (1.0%)	0.008

is the authoritative structure because three of the builders (Erdős–Rényi, small world, scale-free) and the action sampler are unseeded.

Three details of the builders matter for interpretation. First, the small-world, scale-free, echo-chamber, polarised, and hierarchical builders adjust trust for homophily: trust follows the piecewise rule in Section 4.4, retaining the base value for identical persona identifiers, an absent persona map, or an empty tag-set union. Second, the echo-chamber and polarised builders add edges from the seed until its out-degree reaches two, so the seed can never be isolated; the Erdős–Rényi builder has no such guard even though its configuration records the parameter. Third, D-net is not a ninth generator: it is a 63-node, 228-edge custom graph copied from the constructed hashtag co-occurrence object described in Chapter B.

A.5 Declared and realised heterogeneous assignments

The declarations differ from the realised assignments in three clustered conditions, as described in Section 4.3. The complete persona prompts are in `thesisExperiment/personas/merged.json`; the table below records the mix identities and their realised seat rules.

Table A.4: The eight directed eight-node topology generators. The parameters are those recorded in the Phase 2 configuration files; the construction column paraphrases the builder code. Trust values are initial edge weights before relation evolution.

Identifier	Recorded parameters	Construction
linear_chain	edge trust 0.70	Directed edges $v_i \rightarrow v_{i+1}$ for $i = 0, \dots, 6$ (Maurya et al., 2025).
ring	edge trust 0.70	Linear chain plus the closing edge $v_7 \rightarrow v_0$.
random_er	$p = 0.42$; minSeedOutDegree 2 recorded but not read by the builder	Each ordered non-self pair receives an edge with probability p from <code>unseeded Math.random</code> ; trust is $0.3 + U(0, 0.5)$; no seed-degree guard applies (Erdős and Rényi, 1960).
small_world	$k = 4, \beta = 0.10$	Directed ring lattice to two neighbours on each side; each forward-side edge is rewired to a random target with probability β (Watts and Strogatz, 1998).
scale_free	$m = 2$	Fully connected $m + 1$ seed extended by preferential attachment on in-degree; each attachment adds edges in both directions (Barabási and Albert, 1999).
echo_chamber	2 chambers; $p_{\text{in}} = 0.75$, $p_{\text{out}} = 0.08$; trust 0.85 within, 0.15 across; seed out-degree at least 2	Chamber membership alternates by node index; dense high-trust within-chamber and sparse low-trust cross-chamber edges; seeded generator.
polarized	$p_{\text{in}} = 0.75, p_{\text{out}} = 0.05$; trust 0.88 within, 0.10 across; seed out-degree at least 2	Nodes 0–3 and 4–7 form two contiguous halves with sparse low-trust cross edges; seeded generator.
hierarchical	branching factor 3; trust 0.85 downward, 0.30 upward	Rooted ternary tree with parent-to-child edges and weaker child-to-parent edges.

Table A.5: Heterogeneous mixes: declared seat order from `node_0` to `node_7`, declared conspiracy seats, and the realisation under cluster assignment. The sequential generators (chain, ring, Erdős–Rényi, small world, scale-free) seat the declared order; the echo-chamber, polarised, and hierarchical generators seat the realisation given beneath each order, in which every mix except `mix_02` holds four conspiracy seats.

Mix	Declared seat order (<code>node_0</code> to <code>node_7</code>)	Conspiracy seats declared
<code>mix_00</code>	<code>conspiracy_believer</code> , <code>conspiracy_haarp_weather</code> , <code>conspiracy_depopulation</code> , <code>conspiracy_climate_piggyback</code> , <code>climate_action_advocate</code> , <code>climate_justice_youth</code> , <code>mitigation_first_policy</code> , <code>environmental_concern</code> Cluster-assigned realisation: 4 unique personas; 4 conspiracy seats (<code>conspiracy_believer</code> and <code>conspiracy_depopulation</code> alternate at even seats)	4
<code>mix_01</code>	<code>conspiracy_depopulation</code> , <code>conspiracy_climate_piggyback</code> , <code>climate_action_advocate</code> , <code>climate_justice_youth</code> , <code>mitigation_first_policy</code> , <code>environmental_concern</code> , <code>ozone_stratosphere_specialist</code> , <code>biodiversity_food_security</code> Cluster-assigned realisation: 4 unique personas; 4 conspiracy seats (<code>conspiracy_depopulation</code> at all even seats)	2
<code>mix_02</code>	<code>climate_action_advocate</code> , <code>climate_justice_youth</code> , <code>mitigation_first_policy</code> , <code>environmental_concern</code> , <code>ozone_stratosphere_specialist</code> , <code>biodiversity_food_security</code> , <code>climate_scientist</code> , <code>science_journalist</code> Cluster-assigned realisation: 8 unique personas, as declared; 0 conspiracy seats	0
<code>mix_03</code>	<code>mitigation_first_policy</code> , <code>platform_moderator_toxicity</code> , <code>ozone_stratosphere_specialist</code> , <code>climate_action_sweden_scopex</code> , <code>biodiversity_food_security</code> , <code>climate_scientist</code> , <code>conspiracy_believer</code> , <code>climate_justice_youth</code> Cluster-assigned realisation: 5 unique personas; 4 conspiracy seats (<code>conspiracy_believer</code> at all even seats)	1
<code>mix_04</code>	<code>climate_action_advocate</code> , <code>conspiracy_peripheral_skywatcher</code> , <code>science_journalist</code> , <code>biodiversity_food_security</code> , <code>climate_action_sweden_scopex</code> , <code>platform_moderator_toxicity</code> , <code>ozone_stratosphere_specialist</code> , <code>conspiracy_depopulation</code> Cluster-assigned realisation: 4 unique personas; 4 conspiracy seats (<code>conspiracy_depopulation</code> at all even seats)	2
<code>mix_05</code>	<code>climate_scientist</code> , <code>climate_action_advocate</code> , <code>conspiracy_climate_piggyback</code> , <code>biodiversity_food_security</code> , <code>platform_moderator_toxicity</code> , <code>science_journalist</code> , <code>ozone_stratosphere_specialist</code> , <code>conspiracy_peripheral_skywatcher</code> Cluster-assigned realisation: 4 unique personas; 4 conspiracy seats (<code>conspiracy_climate_piggyback</code> at all even seats)	2

Appendix B

D-net Construction and Structural Audit

B.1 Construction and reference object

D-net contains 63 hashtag-defined roles and 228 directed arcs (171 unique undirected pairs). Its edge-generation ledger distinguishes 100 arcs from selected hashtag pairs, 95 from same-cluster completion, 17 from same-hashtag expansion, 13 from chemtrails-hub connections, and three from expert-role connections. OSF counts annotate 28, 13, zero, seven, and zero arcs in these respective categories: 48 annotations in total, with no edge added or removed. The graph is therefore a documented construction rather than an observed user network.

The mixed assignment seats 28 conspiracy, 19 climate-action, 13 environmental-concern, and three expert profiles. Fifty-six nodes carry the exact intended identifier and seven a coarser identifier from the same family. The homogeneous assignment places `conspiracy_believer` everywhere. On unique undirected pairs, transitivity is 0.31 and mean local clustering 0.52. Modularity for the conspiracy-versus-other partition is 0.43; a single homogeneous community has zero modularity by construction. Family homophily is 0.84 in the mixed assignment and 1.00 in the homogeneous assignment.

The reference statistics—depth four, breadth 33, and structural virality 2.70—are computed from the seed’s traversal of this same graph. Depth is the longest breadth-first distance, breadth the largest level, and virality the mean pairwise undirected distance among reached nodes (Goel et al., 2016). The reference reaches 61 nodes and carries no clock or empirical claim-loss scores. It supplies no independent Twitter comparison.

B.2 Extraction defect and corrected rule

The archived extractor used the penultimate provenance entry as a node’s parent. The stored incoming chain ends at the sender, so that entry denotes the sender’s sender. On $A \rightarrow B \rightarrow C$, the rule returns $A \rightarrow C$ and omits $A \rightarrow B$. Its fallback root selection also chose a node other than the seed in three of eight retained cells. Finally, the original comparison pooled 16 cells from two executions of each configuration, including two cells that ended after one event. Their pooled depth, breadth, and virality were 2.56, 40.94, and 1.80; these are superseded diagnostics.

The corrected first-receipt tree attaches each non-seed node to the `sourceNodeId` of its first recorded event, including drops, and fixes the root at `user_chemtrails_hub`. Ties use tick and then history order. Every retained live tree reaches 61 nodes through

Table B.1: Cascade-shape statistics of the eight retained D-net cells (four retained run directories, two articles) under the archived rule as executed (grandparent parent, file-order root), the corrected first-receipt tree (parent = recorded sender of the node’s first event, root = seed), and the sender-graph variant. Entries list depth, breadth, and structural virality, in that order; size is 61 in every cell under every rule. Reference: 4 / 33 / 2.70.

Cell	Archived rule	First-receipt tree	Sender graph
Dnet_c_H_conspiracy (run 2026-09-19_02-49-31)			
scopex_2017	3 / 48 / 2.04	4 / 33 / 3.14	4 / 33 / 2.78
chemtrails_gates_2018_2021	3 / 48 / 2.06	5 / 33 / 3.24	4 / 33 / 2.81
Dnet_c_He_mixed (run 2026-09-19_04-05-50)			
scopex_2017	3 / 47 / 2.09	5 / 33 / 3.23	5 / 33 / 2.94
chemtrails_gates_2018_2021	2 / 32 / 2.04 ^a	4 / 33 / 3.14	4 / 33 / 2.80
Dnet_d_H_conspiracy (run 2026-09-19_02-49-31)			
scopex_2017	3 / 49 / 2.04	4 / 33 / 3.14	4 / 33 / 2.76
chemtrails_gates_2018_2021	3 / 49 / 2.05	4 / 33 / 3.14	4 / 33 / 2.78
Dnet_d_He_mixed (run 2026-09-19_04-07-40)			
scopex_2017	3 / 39 / 2.04 ^a	4 / 33 / 3.14	4 / 33 / 2.78
chemtrails_gates_2018_2021	3 / 50 / 2.05 ^a	4 / 33 / 3.18	4 / 33 / 2.79
Mean of the eight cells	2.88 / 45.25 / 2.05	4.25 / 33.00 / 3.17	4.13 / 33.00 / 2.81
Archived pooled 16 outcomes	2.56 / 40.94 / 1.80	—	—

Source: `notes/dnet_reconstruction.json` (`notes/verify_dnet.py`); archived per-cell values cross-checked against `results_phase2/debnath_compare.json`.

^a Root chosen by file order was not the seed (`user_srm_amp`, `user_haarp_hub`, `user_srm_amp`); with the seed forced as root the archived rule gives 3 / 49 in all three cells.

60 edges. A sensitivity variant takes the union of sender-to-recipient edges on forward and reinterpret events, retaining multiple parents and cycles. Table B.1 distinguishes these objects and reports all eight retained cells.

B.3 What the corrected match establishes

Mean first-receipt depth, breadth, and virality are 4.25, 33.00, and 3.17, versus 4, 33, and 2.70 for the input-graph reference. All eight retained cells fall inside the chosen 30 per cent tolerance on each statistic. The sender-graph means are 4.13, 33.00, and 2.81. These checks show that recorded propagation broadly follows the supplied wiring; a close match is expected when many messages traverse that same graph. Two first-receipt trees reach depth five, so the correspondence is approximate rather than an identity.

The archived composite combined structural similarity (weight 0.40), a Jensen–Shannon distributional term (0.40), and content correlation (0.20), with a pass threshold of 0.70. Content correlation was fixed at zero because no empirical content measure exists. Correcting the first-receipt extraction moves the composite from approximately 0.28 to 0.62, but neither its weights nor its threshold have external validation. Distributional comparisons against one constructed reference cannot establish population fit. Full formulas, archived and corrected outputs, and the 16-cell sensitivity remain in the companion technical record; no substantive conclusion relies on the composite.

Appendix C

Scope, Provenance, and Reproduction

C.1 Proposal-to-execution record

The executed study differs materially from the April 2026 proposal. Planned hydration of the Debnath tweet identifiers could not run: credentials were unavailable, the Mendeley files interface failed, and sampled OSF identifiers had unrecoverable floating-point precision loss. The study instead uses six curated article summaries and author-designed persona families informed by the published discourse themes. The OSF pathway (Debnath, 2022b) yielded hashtag aggregates only.

The proposed empirical-network comparison became a stress test on a constructed hashtag graph. Its nodes are synthetic roles, and the structural reference is derived from its input edges. The thesis therefore withdraws any interpretation as a validated digital twin. The full seven-factor firestorm account was also not implemented as an experimental model: only bundled proxies for selected factors are represented, with no affect or communication-volume outcome.

The campaign retained one run per configuration and an eight-tick horizon for practical resource reasons. The earlier Phase 1 pilot explored article, homogeneous, and heterogeneous conditions and informed Phase 2, but it is not a replication sample or an additional confirmatory result. The prior framework’s 30-hop setting was not reproduced in a matched horizon study. C1–C5 and subsequent archive audits were defined after execution. These departures change the evidential target to a descriptive measurement audit; the record does not establish that each change received prior supervisory approval.

The companion archive preserves the full original proposal-to-execution ledger, pilot appendix, secondary plots, and superseded diagnostics in the pre-tightening manuscript snapshot. Their removal from the printed document does not remove them from the technical record.

C.2 Contribution and source provenance

The auditor–node architecture, MI, MPR, and general-news chain study originate in prior co-authored work (Maurya et al., 2025). The thesis-specific research comprises the climate stimuli and item set, persona assignments, network campaign, measurement and population audits, and interpretation. This component boundary is not a claim that every current engine routine predates the thesis or was written solely by the candidate.

The repository history records later implementation changes: the audited graph builder at commit `c05d99e` and auditor at `ffd876f` date to 18 September 2026; the node implementation at `d18ce01` and structural extractor at `4cc2a5f` date to May 2026; the parser at `6efcca1` dates to 19 September. Full component and persona provenance ledgers remain with the source. Commit identity documents an artefact revision, not independently verified personal authorship.

The public research repository is https://github.com/RajGM/LLM_Society. The earlier cited revision `0420234` does not contain the subsequent reconciliation and verification additions. The companion archive supplied with this version therefore includes those additions, the analysis inputs, retained histories, figure sources, and current manuscript. Its `SHA256SUMS.json` identifies the exact bytes supplied, including the regeneration script. The archive records the research working-tree baseline `69e3c0c` and manuscript baseline `ab7bcfc`; it is not represented as an already published DOI release. Licence and persistent-release arrangements remain to be resolved by the author; the existing README and package declarations are inconsistent.

C.3 Two incomplete event histories

Reparsing the fixed run manifest reproduces 1,739 of 1,741 historical rows. Two `polar_bears` cells differ: `T2d_H_polarized_ozone_stratosphere_specialist` and `T2d_H_scale_free_biodiversity_food_security`. Their archived node histories were captured during auditing, with 37 of 122 and 18 of 153 ultimately scored events present. Later aggregate summaries corroborate the completed historical values, but the missing 85 and 135 event-level scores were never archived.

The thesis uses the reconciled event-based means: 1.5405 rather than 1.4836 for the polarised cell, and 0.8333 rather than 1.0784 for the scale-free cell. Both remain live; maximum scores and k^* are unchanged. Some rounded summaries and ranks change, including the cross-campaign H rank correlation from 0.10 to -0.05 and the same-event H correlation from 0.67 to 0.62. The full field-level differences remain in `thesisExperiment/results_phase2/tables_reconciled/reconciliation_diff.json`.

The fixed manifest names 297 run directories and 1,741 rows, including the 288-configuration, 1,728-cell main grid and separate auxiliary records. It replaces modification-time selection, which changes nine selections after a fresh checkout. Regeneration uses the parser's original cell and CSV functions through a read-only filesystem interface, with this manifest held fixed. Rebuilding the manifest or running the historical parser's selection routine is not part of reproduction.

C.4 Reproduction from the companion archive

Extract the companion archive and run commands from its root. The archive contains `reproduce.py`, a driver that requires Node.js and Python with the versions recorded in `6aaeb12d2d2eaea03e67a80b/notes/requirements-analysis.txt`. It regenerates the reconciled CSV files in an isolated output directory, checks them against the supplied

tables, and runs the numerical analysis and verification scripts without calling a model API. The fixed source inputs are preserved.

```
python -m pip install -r 6aaeb12d2d2eaea03e67a80b/notes/requirements-  
analysis.txt  
python reproduce.py
```

Listing C.1: Offline numerical verification from the extracted archive root.

The driver writes a verification report under `reproduction_output/`. The figure builder and build instructions are also supplied; compiling the manuscript uses `6aaeb12d2d2eaea03e67a80b/build.ps1` on Windows or `latexmk -pdf main.tex` inside that directory. Numerical reproducibility is the claim: PDF metadata and library rendering may change file bytes. Exact replay of the model requests is unsupported because generation randomness and hosted-model state were not archived sufficiently.

Appendix D

Robustness and Observation-Window Tables

This appendix retains the supplementary evidence behind Section 5.4.4: the robustness of every principal finding to the population of scored cells, the extent to which the eight-tick window truncated the observed cascades, and the deterministic inbox-admission rule that decided which deliveries entered a node’s inbox. `notes/verify_supplementary.py` computed all three read-only from the retained run directories named in the run manifest (Section C.3), recorded the numbers in `notes/verify_supplementary.json`, and generated the supplementary data used below. Cell means, k^* fields, and live status derive from the reconciled table set; event-level quantities (ticks, rewrite depth, deliveries) derive from the archived node histories.

D.1 Robustness to scoring coverage

The principal comparisons summarised in Table D.1 were recomputed on the full live-cell population and two restricted subsets. Population (i) comprises all 1,436 live cells (480, 238, 478, and 240 in T2c H, T2c He, T2d H, and T2d He). Population (ii) comprises the 840 live cells of the 167 runs whose run log records no HTTP error (`api_error_by_run.csv`, `http_errors` equal to zero); it contains no dual-discrete He cell for the scale-free, echo-chamber, or polarised conditions, so its dual-discrete He entries rest on five conditions. For the dual-discrete He same-event rank comparison these contribute 139 cells (chain 33, ring 33, Erdős–Rényi 29, small world 16, hierarchical 28). The continuous He subset instead contains 150 cells across seven conditions, including three polarised and eight scale-free cells; only the echo chamber is absent.

Population (iii) comprises the 1,125 live cells in 245 runs in which every audit-eligible event received a score (`nScored` equal to the number of eligible events in the raw history); identity rewrites, whose output equals their input, are not excluded. The two restricted subsets overlap in 838 cells: two cells from runs without HTTP errors are not fully scored, and 287 fully scored cells belong to runs with an HTTP error. Neither restricted subset therefore contains the other. Table D.1 lists each finding on the three populations.

The main family and mix contrasts retained their direction across the three populations (Table D.1). Conspiracy scored above climate action, which scored above science and environment, and `mix_00` remained above `mix_02` under both instruments. The dual-discrete He mean remained above H, whereas the continuous He-minus-H difference changed sign in the zero-HTTP-error subset (-0.03). The He endpoint-rate reversal between instruments also remained.

Table D.1: Robustness of the principal findings to the population of scored cells: all live cells, live cells of zero-HTTP-error runs, and live cells with every eligible event scored. Numbers in parentheses are live-cell counts. The zero-HTTP-error population contains no dual-discrete He cell for the scale-free, echo-chamber, or polarised conditions.

Finding	All live cells	Zero-error runs	Fully scored cells
<i>Live cells in subset</i>			
Live cells (T2c H / T2c He / T2d H / T2d He)	480/238/478/240	247/150/304/139	355/186/402/182
<i>Family live-cell mean MI (H arm; continuous / dual-discrete)</i>			
Conspiracy	2.22 (168) / 3.17 (159)	2.23 (93) / 3.17 (107)	2.20 (122) / 3.15 (132)
Climate action	0.17 (117) / 1.05 (126)	0.24 (50) / 1.13 (75)	0.20 (85) / 1.05 (119)
Science and environment	0.11 (195) / 0.87 (193)	0.14 (104) / 0.93 (122)	0.12 (148) / 0.87 (151)
Conspiracy highest	yes / yes	yes / yes	yes / yes
<i>Mix contrast (He arm; continuous / dual-discrete)</i>			
mix_00	1.77 (39) / 3.13 (40)	1.70 (29) / 3.01 (26)	1.68 (37) / 3.03 (32)
mix_02	0.26 (41) / 1.19 (41)	0.30 (26) / 1.29 (22)	0.28 (32) / 1.33 (31)
mix_00 > mix_02	yes / yes	yes / yes	yes / yes
<i>Article ordering (lowest / highest article per slice)</i>			
T2c H	polar bears / consensus	polar bears / consensus	polar bears / consensus
T2c He	polar bears / consensus	polar bears / chemtrails	polar bears / chemtrails
T2d H	polar bears / consensus	polar bears / consensus	polar bears / consensus
T2d He	SAI / consensus	polar bears / consensus	polar bears / consensus
<i>Within-window k^* occurrence among live cells (%)</i>			
T2c H	14.2 (480)	15.0 (247)	14.1 (355)
T2c He	7.6 (238)	6.7 (150)	6.5 (186)
T2d H	20.1 (478)	21.4 (304)	19.9 (402)
T2d He	32.1 (240)	35.2 (139)	33.5 (182)
<i>Same-event condition-mean rank correlation, dual cells (Spearman, available conditions)</i>			
H	0.62 (478)	0.90 (304)	0.95 (402)
He	0.12 (240)	0.70 (139) ^a	0.83 (182)
<i>Pooled He – H (live-cell means; continuous / dual-discrete)</i>			
He – H	+0.15 / +0.40	–0.03 / +0.47	+0.13 / +0.48

Source: `notes/verify_supplementary.py` on the reconciled table set and the retained node histories; body in `notes/robustness_table.tex`. The H same-event rank correlation on all live cells is 0.62 under the reconciled set (0.67 under the historical table, Section C.3); no other entry differs at two decimals between the two table sets.

^a Five conditions only (no dual-discrete He cell of the scale-free, echo-chamber, or polarised conditions belongs to a run without HTTP errors).

Table D.2: What a defined within-window k^* observes, by slice and scored series: live and crossing cells, crossing cells with 0 (terminal only), 1, and 2 or more scored ticks after k^* , live cells with one scored tick, and cells whose last scored tick precedes their last audit-eligible tick (live / crossing). A cell has a defined k^* if and only if mean MI at its last scored tick exceeds 3 (0 violations in 2,154 checks).

Slice	Series	Live	k^* defined		Scored ticks after k^*			Coverage	
			n	%	0 (terminal only)	1	≥ 2	One scored tick (live)	Gap (live / k^*)
T2c H	continuous	480	68	14.2	7 (10.3 %)	9	52	4	28 / 3
T2c He	continuous	238	18	7.6	3 (16.7 %)	5	10	1	9 / 0
T2d H	discrete headline	478	96	20.1	5 (5.2 %)	10	81	5	28 / 7
T2d H	continuous sidecar	478	61	12.8	10 (16.4 %)	6	45	5	28 / 5
T2d He	discrete headline	240	77	32.1	14 (18.2 %)	18	45	0	19 / 3
T2d He	continuous sidecar	240	22	9.2	5 (22.7 %)	7	10	0	19 / 0

Article ordering and same-event rank agreement were more sensitive to the selected population. The lowest article in full-population T2d He was SAI rather than polar bears; in both restricted populations, chemtrails replaced consensus as the highest article in T2c He. Same-event rank correlations rose from 0.62 (H) and 0.12 (He) to 0.90 and 0.70 in the zero-HTTP-error subset, and to 0.95 and 0.83 among fully scored cells. The former subset excludes three dual-discrete He network conditions, so these changes cannot isolate missing-score selection from changes in condition composition. Both restrictions describe selected subsets rather than corrected full-population estimates. Fully scored cells can still contain failed rewrites whose unchanged-text fallbacks suppress either distortion or correction.

The minimum-score cutoff changes magnitudes without reversing the pooled arm directions at the tested values. Requiring at least one, two, three, or five scored events per cell gave He-minus-H differences of 0.170, 0.155, 0.142, and 0.088 under continuous scoring, and 0.386, 0.399, 0.416, and 0.456 under dual-discrete scoring. Including the 60 single-score cells also retained the between-instrument reversal of the He endpoint rate. These are comparisons of different included populations, not independent replications. The companion script `notes/verify_review_checks.py` records their counts and endpoint rates.

D.2 Threshold endpoints

Table D.2 records how much follow-up the above-threshold suffix contains and distinguishes the discrete headline from the continuous sidecar on the same events.

D.3 Observation window

The eight-tick window truncated most live cascades rather than observing their end. Of the 1,436 live cells, 1,334 (92.9 per cent) recorded at least one event at tick 8 and 91.5

per cent recorded an audit-eligible event at tick 8; by slice the shares were 94.0, 87.4, 95.6, and 90.8 per cent for T2c H, T2c He, T2d H, and T2d He. Every live cell of the small-world, scale-free, echo-chamber, and polarised conditions was still producing events at tick 8, as were 98.8 per cent of Erdős–Rényi cells, 86.7 per cent of hierarchical cells, 80.5 per cent of chain cells, and 79.2 per cent of ring cells; the two path-like conditions stop early only when a drop empties the queue. The last scored tick was tick 8 in 1,230 cells (85.7 per cent). Of the 104 cells active but unscored at tick 8, 20 recorded only drops and 84 recorded audit-eligible events without a score.

The window nevertheless contained 43.6 text-changing rewrites per live cell on average (median 44, range 0 to 113; the raw `reinterpret` label counts 52.2 per cell because it includes identity fallbacks whose output equals their input); by condition the means were 6.4 (chain), 6.3 (ring), 19.5 (hierarchical), 45.0 (polarised), 60.1 (echo chamber), 68.9 (Erdős–Rényi), 70.4 (small world), and 77.8 (scale-free). Rewrite depth counts the text-changing rewrites in a scored text’s lineage. Reconstruction uses the recorded sender at the previous tick, exact matching of input and output texts, and incoming provenance; all 153,855 events have a resolved, unambiguous compatible depth in this archive. The per-cell maximum was at least 6 in 1,009 cells and 8 in 227. Across the 1,436 live cells, the median maximum lineage depth was six sequential transformations. This remains below the 30 hops of the prior chain study.

At the four tested truncation points, the pooled means continued to change while the family ordering remained the same. Figure D.1 plots the live-cell means computed from events with tick at most T for $T = 2, 4, 6, 8$. The T2d H mean rose from 1.37 to 1.57, 1.64, and 1.68, with successive increments of +0.19, +0.07, and +0.04; the T2c H mean rose from 0.61 to 0.75, 0.84, and 0.86; the He means rose from 0.83 to 1.02 (T2c) and from 1.64 to 2.08 (T2d). Under the continuous instrument the conspiracy family rose from 1.62 to 2.22, and under the dual-discrete instrument from 2.47 to 2.92, 3.11, and 3.17, so its mean crossed 3 only once the window included ticks 5 and 6 ($T = 6$, compared with $T = 4$). The family ordering, conspiracy above climate action above science and environment, was the same at every truncation point under the continuous, dual-discrete, and sidecar readings. The window is therefore a bounded early-propagation audit: the pooled means still changed between $T = 6$ and $T = 8$, while the family ordering in Section 5.3 held at the tested windows. Neither pattern establishes convergence beyond tick 8 or convergence of individual cells.

D.4 Inbox admission

The inbox cap of four is a deterministic exposure policy, not random thinning (Section 4.5.1). Within a tick the engine processes nodes in stored order (`node_0` first), appends each node’s outgoing messages to one delivery list, and applies the deliveries in list order; a recipient’s inbox admits a message only while it holds fewer than four queued messages for the same article, and the engine discards any further delivery without an event, counter, or log line. Reconstruction from the event record confirms the rule exactly: for every one of the 49,817 recipient–tick slots, the number of accepted deliveries

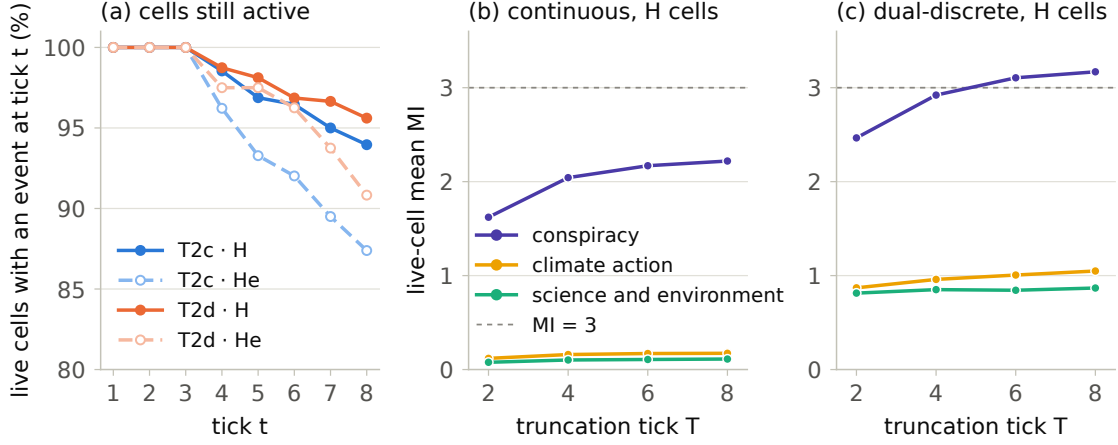


Figure D.1: Observation-window characterisation. (a) Share of live cells with at least one event at tick t for each campaign slice; 92.9 per cent of the 1,436 live cells were still active at tick 8. (b) and (c) Live-cell mean MI of homogeneous cells by persona family when only scored events with tick $\leq T$ are used, for $T = 2, 4, 6, 8$, under the continuous instrument (T2c) and the dual-discrete headline (T2d); the dashed line marks $MI = 3$. Denominators increase with T because a cell enters only once it has a scored event within the window (conspiracy H: 133 to 168 cells in T2c, 138 to 159 in T2d). Longer windows add both observations and eligible cells; the family ordering is identical at every tested truncation point under both instruments.

equalled the smaller of four and the number proposed, and in every one of the 27,777 capacity-bound slots the accepted senders were the first four in node order (0 violations). Proposed deliveries to a recipient at tick $t + 1$ are the forward and reinterpret events at tick t of nodes with an edge to it; accepted deliveries are the events the recipient recorded at tick $t + 1$.

The rule discarded more than half of all proposed deliveries, with rejection rates increasing monotonically with the sender’s seat number. Capacity bound in 27,777 of 49,817 recipient–tick slots (55.8 per cent; 55.8, 56.6, 56.1, and 54.2 per cent by slice), and 172,453 of 324,580 proposed deliveries (53.1 per cent) were discarded, none in the chain or ring conditions, where every node has at most one incoming edge (Table D.3a and b). The seed’s output was never discarded, whereas `node_7` lost 88.8 per cent of its deliveries overall and 98.0 per cent in the scale-free condition; the rejection rate rose monotonically with the sender identifier from 0.0 per cent (`node_0`) through 20.9, 46.7, 67.0, 74.0, 76.1, and 86.8 per cent to 88.8 per cent (`node_7`). By condition the discarded share ranged from 25.7 per cent (hierarchical) through 45.2 (polarised), 49.2 (Erdős–Rényi), 51.3 (echo chamber), and 57.1 (scale-free) to 63.1 per cent (small world).

Sender family and node order are associated under the realised heterogeneous assignments. Conspiracy senders lost less exposure than science and environment senders in each dense condition listed in Table D.3c. This pattern reflects the interaction of placement with deterministic admission, not an isolated family effect. Randomised or counterbalanced delivery order and explicit rejection logs are required to separate these mechanisms.

Table D.3: Inbox-admission outcomes reconstructed from the event record of the 288 retained grid runs. (a) Proposed and rejected deliveries by sender node identifier, all slices. (b) Recipient-tick slots, share of slots at capacity, and rejected deliveries by network condition. (c) Rejection rate by sender persona family in the heterogeneous arm for the six non-path conditions.

<i>(a) By sender node identifier</i>			
Sender	Proposed	Rejected	Rejected (%)
node_0 (seed)	50,307	0	0.0
node_1	43,335	9,069	20.9
node_2	52,280	24,416	46.7
node_3	40,707	27,260	67.0
node_4	37,431	27,716	74.0
node_5	35,795	27,235	76.1
node_6	35,754	31,030	86.8
node_7	28,971	25,727	88.8
All senders	324,580	172,453	53.1

<i>(b) By network condition</i>					
Condition	Slots	At capacity (%)	Proposed	Rejected	Rejected (%)
Linear chain	1,268	0.0	1,268	0	0.0
Ring	1,229	0.0	1,229	0	0.0
Hierarchical	4,126	18.1	13,547	3,477	25.7
Polarised	7,946	54.0	44,428	20,093	45.2
Erdős-Rényi	7,609	52.8	45,967	22,617	49.2
Echo chamber	9,104	61.0	60,147	30,835	51.3
Scale-free	9,174	69.8	71,732	40,990	57.1
Small world	9,361	72.2	86,262	54,441	63.1
All conditions	49,817	55.8	324,580	172,453	53.1

<i>(c) He arm, rejection rate (%) by sender persona family</i>			
Condition	Conspiracy	Science and environment	Climate action
Hierarchical	0.0	— ^a	1.2
Polarised	44.9	60.9	24.3
Erdős-Rényi	36.6	69.1	40.3
Echo chamber	46.6	62.1	19.9
Scale-free	41.8	76.3	43.0
Small world	48.1	77.5	51.3

Source: `notes/verify_supplementary.py` (item M16) from `graph_topology.json` and `nodes/*.json` of the retained runs; a slot is one recipient, one tick, one article. Chain and ring never reach capacity because each node has at most one incoming edge. Mix-only personas are omitted from (c).

^a The reconstructed hierarchical He events contain no proposed outgoing deliveries from the science-and-environment sender group, so its rejection rate is undefined. This is a delivery count, not a count of assigned persona seats.

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Any remaining errors are my own.

Declaration

I hereby declare that this thesis is entirely the result of my own work except where otherwise indicated. I have only used the resources given in the list of references.

Declaration of Use of Generative AI

Category	GenAI Use
Idea Generation & Concept Development	None
Literature Search and Analysis	None
Methodology	None
Coding	Cursor Composer (code writing, debugging)
Data Collection	GPT-4o-mini (experimental instrument)
Data Analysis	Cursor Composer (analysis scripts)
Interpretation and Validation	None
Structuring and Planning of the Text	None
Text Generation	None
Translation of Text	N/A
Proofreading and Editing of Text	Grammarly (grammar, spelling)
Image Generation	None
Data Visualisation	None
Presentation	None
Reference List	None
Accessibility	N/A

I hereby declare that this thesis is entirely the result of my own work except where otherwise indicated. I have only used the resources given in the list of references. The following tools were used during the preparation of this thesis: Cursor Composer (AI-assisted code editor) for writing and debugging the simulation software; Grammarly for grammar and spelling corrections; and OpenAI GPT-4o-mini as the large language model powering the simulated agents, which constitutes the experimental instrument under study. The research design, methodology, analysis, interpretation of results, all written text, and the responsibility for every statement in this thesis are my own.

Munich, 21-09-2026
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